# Data Center Facility Specifications

Meridian Digital Solutions
Facilities Management Division

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# 1. Executive Facility Overview

Meridian Digital Solutions operates three geographically distributed, enterprise-grade colocation data center facilities designed to support high-availability hosting, regional failover capabilities, disaster recovery standby systems, and software development testing environments. The current facility portfolio consists of DC-East in Ashburn, Virginia (the primary production facility), DC-West in The Dalles, Oregon (used for disaster recovery and secondary regional production hosting), and DC-Central in Elk Grove Village, Illinois (serving as a Midwest regional hub and secondary production site). Combined, these facilities house 147 active physical and virtual servers across 195 populated racks, drawing a total electrical load of approximately 3.55 MW against a combined utility capacity of 5.2 MW (representing a 68% total capacity utilization across all sites). The total annual colocation lease expenditure across all three sites is $852,000, representing a significant portion of the corporate IT operations budget. The facilities are managed by the Facilities Management Division in coordination with the Infrastructure Operations and Network Engineering teams to ensure 99.999% uptime for power, cooling, and network carrier availability.

Key Strategic and Operational Constraints:

- DC-Central Lease Expiration: The lease expires on June 30, 2025, creating an urgent operational timeline to migrate all 48 active servers to the other sites before penalty clauses or lease termination occurs.
- DC-West Lease Renewal: The lease expires in September 2025. Negotiations are active, but power rate escalations along the Columbia River are projected to increase operating costs by 12% post-renewal.
- DC-East Facility Runway: The facility has the longest operational runway, with a lease secured through March 31, 2028. This long-term stability makes it the logical target for the primary consolidated site.
- Consolidation Budget: Project CAPEX-2025-001 has been approved with a total capital budget of $1,100,000 covering relocation logistics, temporary parallel bandwidth, and hardware refreshes for the target site.
- Target Consolidated State: The strategic objective is to consolidate from three sites to two sites, leveraging DC-East as the primary hosting facility and DC-West as the active-passive disaster recovery standby facility, supplemented by AWS public cloud extensions for burstable workloads.

The following sections provide detailed technical specifications for each colocation facility, including utility power feeds, cooling system redundancy, telecommunication carrier density, lease terms, and rack-level power distribution metrics.

# 2. DC-East - Ashburn, Virginia

The DC-East facility is located in the Loudoun County Technology Corridor in Ashburn, Virginia, the largest data center market globally. This facility serves as the primary production data center and hosts the majority of Tier 1 mission-critical applications. The facility is operated by QTS Data Centers with a dedicated cage and private suite for Meridian.

| | |
|---|---|
| **Facility Name** | DC-East |
| **Street Address** | 21715 Filigree Court, Ashburn, VA 20147 |
| **Colocation Provider** | QTS Data Centers (QTS-ASH-1) |
| **Total Floor Space** | 12,000 sq ft (raised floor: 10,500 sq ft) |
| **Total Rack Positions** | 120 racks (42U standard) |
| **Racks in Use** | 95 racks (79% utilized) |
| **Power - Utility Feeds** | Dual redundant utility feeds from Dominion Energy |
| **Power - Generator Capacity** | 3.0 MW diesel generator (Caterpillar C175-16) |
| **Power - Generator Fuel** | 48-hour runtime at full load (10,000 gal tank) |
| **Power - UPS Capacity** | 2.4 MW UPS (Eaton 93PM, 2N redundant) |
| **Power - Current Draw** | 1.8 MW (75% of UPS capacity) |
| **Power - PUE** | 1.45 (trailing 12-month average) |
| **Power - Cost** | $0.085/kWh (Dominion Energy commercial rate) |
| **Cooling - CRAC Units** | 8 x Liebert CRV 35kW (N+1 configuration) |
| **Cooling - Total Tonnage** | 80 tons refrigeration capacity |
| **Cooling - Containment** | Hot aisle containment (all rows) |
| **Cooling - Redundancy** | N+1 (7 required, 8 installed) |
| **Network - Carrier Diversity** | 3 carriers: Zayo, Lumen, Crown Castle |
| **Network - Fiber Paths** | Dual diverse fiber paths (separate conduits) |
| **Network - IX Connectivity** | Equinix Ashburn IX (direct peering) |
| **Network - WAN Bandwidth** | 10 Gbps MPLS + 2 Gbps Internet DIA |
| **Lease - Start Date** | March 1, 2018 |
| **Lease - End Date** | March 31, 2028 |
| **Lease - Renewal Options** | 5-year extension to March 2033 |
| **Lease - Monthly Cost** | $35,000 (base) + power pass-through |
| **Physical Security** | Biometric + badge, 24/7 NOC, mantrap, CCTV (90-day) |
| **Certifications** | SOC 2 Type II, ISO 27001, PCI-DSS, SSAE-18 |
| **Server Count** | 55 total (53 active, 2 decommissioned) |

## 2.1 DC-East Rack Power Density

| Rack ID | Row | Alloc kW | Draw kW | Util % | Servers | Primary App |
|---|---|---|---|---|---|---|
| RE-01 | A | 8.0 | 6.2 | 78% | 4 | ERP-Core, CRM-Enterprise |
| RE-02 | A | 8.0 | 5.8 | 73% | 4 | FinanceHub, PayrollEngine |
| RE-03 | A | 10.0 | 8.1 | 81% | 3 | DataWarehouse (high-density) |
| RE-04 | B | 8.0 | 6.5 | 81% | 4 | Email-Exchange cluster |
| RE-05 | B | 8.0 | 5.2 | 65% | 4 | SharePoint-Intranet, HR-Portal |
| RE-06 | B | 8.0 | 7.1 | 89% | 4 | SecuritySIEM, MonitoringDashboard |
| RE-07 | C | 10.0 | 9.2 | 92% | 3 | E-Commerce-Platform (high-density) |
| RE-08 | C | 8.0 | 4.8 | 60% | 4 | API-Gateway, VPN-Gateway |
| RE-09 | C | 8.0 | 5.5 | 69% | 4 | CustomerPortal, ComplianceTracker |
| RE-10 | D | 8.0 | 6.0 | 75% | 4 | LDAP-Directory, DNS-Primary |

| | | | | | | |
|---|---|---|---|---|---|---|
| RE-11 | D | 8.0 | 6.8 | 85% | 4 | BI-Analytics, SupplyChainPro (backup) |
| RE-12 | D | 6.0 | 3.2 | 53% | 3 | Development servers |
| RE-13 | E | 8.0 | 5.9 | 74% | 4 | CI-CD-Pipeline, DevOps-Pipeline |
| RE-14 | E | 8.0 | 7.4 | 93% | 3 | VMware vCenter, management infra |
| RE-15 | E | 6.0 | 2.8 | 47% | 2 | Backup infrastructure |
| RE-16 | F | 10.0 | 8.5 | 85% | 4 | Emergency provisioned servers |
| RE-17 | F | 8.0 | 5.1 | 64% | 3 | Network appliances (Palo Alto, F5) |
| RE-18 | F | 6.0 | 3.9 | 65% | 3 | Storage arrays (NetApp AFF) |
| RE-19 | G | 8.0 | 5.5 | 69% | 4 | Mixed workloads |
| RE-20 | G | 6.0 | 2.4 | 40% | 2 | Spare capacity |
| RE-21 | B | 6.0 | 4.0 | 66% | 2 | DNS-Primary secondary infrastructure |
| RE-22 | B | 8.0 | 5.4 | 67% | 3 | LDAP-Directory secondary infrastructure |
| RE-23 | B | 8.0 | 5.4 | 67% | 4 | FileServer-Central secondary infrastructure |
| RE-24 | B | 6.0 | 4.1 | 68% | 1 | BackupManager secondary infrastructure |
| RE-25 | B | 8.0 | 5.6 | 70% | 2 | MonitoringDashboard secondary infrastructure |
| RE-26 | B | 8.0 | 5.7 | 71% | 3 | SupplyChainPro secondary infrastructure |
| RE-27 | B | 6.0 | 4.3 | 71% | 4 | MarketingAutomation secondary infrastructure |
| RE-28 | B | 8.0 | 5.8 | 72% | 1 | SalesForce-Integration secondary infrastructure |
| RE-29 | C | 8.0 | 5.9 | 73% | 2 | MobileApp-Backend secondary infrastructure |
| RE-30 | C | 10.0 | 7.5 | 75% | 3 | DocumentMgmt secondary infrastructure |
| RE-31 | C | 8.0 | 6.1 | 76% | 4 | VideoConference secondary infrastructure |
| RE-32 | C | 8.0 | 6.2 | 77% | 1 | ProjectTracker secondary infrastructure |
| RE-33 | C | 6.0 | 4.7 | 78% | 2 | QA-TestSuite secondary infrastructure |
| RE-34 | C | 8.0 | 6.3 | 78% | 3 | LogAggregator secondary infrastructure |
| RE-35 | C | 8.0 | 6.4 | 80% | 4 | Microservice-036 secondary infrastructure |
| RE-36 | C | 6.0 | 4.9 | 81% | 1 | Microservice-037 secondary infrastructure |
| RE-37 | C | 8.0 | 6.6 | 82% | 2 | Microservice-038 secondary infrastructure |
| RE-38 | C | 8.0 | 6.6 | 82% | 3 | Microservice-039 secondary infrastructure |
| RE-39 | C | 6.0 | 5.0 | 83% | 4 | Microservice-040 secondary infrastructure |
| RE-40 | C | 8.0 | 6.8 | 85% | 1 | Microservice-041 secondary infrastructure |
| RE-41 | C | 8.0 | 6.9 | 86% | 2 | Microservice-042 secondary infrastructure |
| RE-42 | C | 6.0 | 5.2 | 86% | 3 | Microservice-043 secondary infrastructure |
| RE-43 | D | 8.0 | 7.0 | 87% | 4 | Microservice-044 secondary infrastructure |
| RE-44 | D | 8.0 | 7.1 | 88% | 1 | Microservice-045 secondary infrastructure |
| RE-45 | D | 10.0 | 9.0 | 90% | 2 | Microservice-046 secondary infrastructure |
| RE-46 | D | 8.0 | 7.3 | 91% | 3 | Microservice-047 secondary infrastructure |
| RE-47 | D | 8.0 | 7.4 | 92% | 4 | Microservice-048 secondary infrastructure |
| RE-48 | D | 6.0 | 5.6 | 93% | 1 | Microservice-049 secondary infrastructure |
| RE-49 | D | 8.0 | 7.5 | 93% | 2 | Microservice-050 secondary infrastructure |
| RE-50 | D | 8.0 | 3.6 | 45% | 3 | Microservice-051 secondary infrastructure |
| RE-51 | D | 6.0 | 2.8 | 46% | 4 | Microservice-052 secondary infrastructure |
| RE-52 | D | 8.0 | 3.8 | 47% | 1 | Microservice-053 secondary infrastructure |
| RE-53 | D | 8.0 | 3.8 | 47% | 2 | Microservice-054 secondary infrastructure |
| RE-54 | D | 6.0 | 2.9 | 48% | 3 | Microservice-055 secondary infrastructure |
| RE-55 | D | 8.0 | 4.0 | 50% | 4 | Microservice-056 secondary infrastructure |
| RE-56 | D | 8.0 | 4.1 | 51% | 1 | Microservice-057 secondary infrastructure |
| RE-57 | E | 6.0 | 3.1 | 51% | 2 | Microservice-058 secondary infrastructure |
| RE-58 | E | 8.0 | 4.2 | 52% | 3 | Microservice-059 secondary infrastructure |
| RE-59 | E | 8.0 | 4.3 | 53% | 4 | Microservice-060 secondary infrastructure |
| RE-60 | E | 10.0 | 5.5 | 55% | 1 | Microservice-061 secondary infrastructure |
| RE-61 | E | 8.0 | 4.5 | 56% | 2 | Microservice-062 secondary infrastructure |

| | | | | | | |
|---|---|---|---|---|---|---|
| RE-62 | E | 8.0 | 4.6 | 57% | 3 | Microservice-063 secondary infrastructure |
| RE-63 | E | 6.0 | 3.5 | 58% | 4 | Microservice-064 secondary infrastructure |
| RE-64 | E | 8.0 | 4.7 | 58% | 1 | Microservice-065 secondary infrastructure |
| RE-65 | E | 8.0 | 4.8 | 60% | 2 | Microservice-066 secondary infrastructure |
| RE-66 | E | 6.0 | 3.7 | 61% | 3 | Microservice-067 secondary infrastructure |
| RE-67 | E | 8.0 | 5.0 | 62% | 4 | Microservice-068 secondary infrastructure |
| RE-68 | E | 8.0 | 5.0 | 62% | 1 | Microservice-069 secondary infrastructure |
| RE-69 | E | 6.0 | 3.8 | 63% | 2 | Microservice-070 secondary infrastructure |
| RE-70 | E | 8.0 | 5.2 | 65% | 3 | Microservice-071 secondary infrastructure |
| RE-71 | F | 8.0 | 5.3 | 66% | 4 | Microservice-072 secondary infrastructure |
| RE-72 | F | 6.0 | 4.0 | 66% | 1 | Microservice-073 secondary infrastructure |
| RE-73 | F | 8.0 | 5.4 | 67% | 2 | Microservice-074 secondary infrastructure |
| RE-74 | F | 8.0 | 5.5 | 68% | 3 | Microservice-075 secondary infrastructure |
| RE-75 | F | 10.0 | 7.0 | 70% | 4 | Microservice-076 secondary infrastructure |
| RE-76 | F | 8.0 | 5.7 | 71% | 1 | Microservice-077 secondary infrastructure |
| RE-77 | F | 8.0 | 5.8 | 72% | 2 | Microservice-078 secondary infrastructure |
| RE-78 | F | 6.0 | 4.4 | 73% | 3 | Microservice-079 secondary infrastructure |
| RE-79 | F | 8.0 | 5.9 | 73% | 4 | Microservice-080 secondary infrastructure |
| RE-80 | F | 8.0 | 6.0 | 75% | 1 | ERP-Core secondary infrastructure |
| RE-81 | F | 6.0 | 4.6 | 76% | 2 | CRM-Enterprise secondary infrastructure |
| RE-82 | F | 8.0 | 6.2 | 77% | 3 | HR-Portal secondary infrastructure |
| RE-83 | F | 8.0 | 6.2 | 77% | 4 | DevOps-Pipeline secondary infrastructure |
| RE-84 | F | 6.0 | 4.7 | 78% | 1 | FinanceHub secondary infrastructure |
| RE-85 | G | 8.0 | 6.4 | 80% | 2 | DataWarehouse secondary infrastructure |
| RE-86 | G | 8.0 | 6.5 | 81% | 3 | CI-CD-Pipeline secondary infrastructure |
| RE-87 | G | 6.0 | 4.9 | 81% | 4 | Email-Exchange secondary infrastructure |
| RE-88 | G | 8.0 | 6.6 | 82% | 1 | SharePoint-Intranet secondary infrastructure |
| RE-89 | G | 8.0 | 6.7 | 83% | 2 | CustomerPortal secondary infrastructure |
| RE-90 | G | 10.0 | 8.5 | 85% | 3 | InventoryMgmt secondary infrastructure |
| RE-91 | G | 8.0 | 6.9 | 86% | 4 | PayrollEngine secondary infrastructure |
| RE-92 | G | 8.0 | 7.0 | 87% | 1 | TravelBooking secondary infrastructure |
| RE-93 | G | 6.0 | 5.3 | 88% | 2 | E-Commerce-Platform secondary infrastructure |
| RE-94 | G | 8.0 | 7.1 | 88% | 3 | BI-Analytics secondary infrastructure |
| RE-95 | G | 8.0 | 7.2 | 90% | 4 | IT-ServiceDesk secondary infrastructure |

# 3. DC-West - The Dalles, Oregon

The DC-West facility is located in The Dalles, Oregon, selected for its proximity to hydroelectric power generation along the Columbia River. This facility provides the lowest operating costs of the three locations and serves as the primary disaster recovery site for Tier 1 and Tier 2 applications. The facility also hosts production workloads for the western region.

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|---|---|
| **Facility Name** | DC-West |
| **Street Address** | 3700 Klindt Drive, The Dalles, OR 97058 |
| **Colocation Provider** | Flexential (FLEX-DAL-2) |
| **Total Floor Space** | 8,000 sq ft (raised floor: 7,200 sq ft) |
| **Total Rack Positions** | 80 racks (42U standard) |
| **Racks in Use** | 52 racks (65% utilized) |
| **Power - Utility Feeds** | Dual feeds from Northern Wasco County PUD |
| **Power - Generator Capacity** | 2.0 MW diesel generator (Cummins QSK60) |
| **Power - Generator Fuel** | 24-hour runtime at full load (5,000 gal) |
| **Power - UPS Capacity** | 1.6 MW UPS (Vertiv Liebert EXL S1, N+1) |
| **Power - Current Draw** | 0.9 MW (56% of UPS capacity) |
| **Power - PUE** | 1.35 (free cooling benefit from climate) |
| **Power - Cost** | $0.045/kWh (hydroelectric rate) |
| **Cooling - Primary** | Free cooling (outside air economizer, 9 months/year) |
| **Cooling - CRAC Backup** | 6 x Liebert CRV 25kW (N+1 for summer months) |
| **Cooling - Total Tonnage** | 55 tons refrigeration capacity |
| **Cooling - Containment** | Hot aisle containment (all rows) |
| **Network - Carrier Diversity** | 2 carriers: Zayo, Wave Broadband |
| **Network - Fiber Paths** | Dual diverse fiber paths |
| **Network - Cloud Peering** | Direct peering with AWS us-west-2 (1 Gbps) |
| **Network - WAN Bandwidth** | 5 Gbps MPLS + 1 Gbps Internet DIA |
| **Lease - Start Date** | September 1, 2020 |
| **Lease - End Date** | September 30, 2025 |
| **Lease - Renewal Options** | 3-year extension under negotiation |
| **Lease - Monthly Cost** | $18,500 (base) + power pass-through |
| **Physical Security** | Biometric + badge, 24/7 security guard, CCTV (60-day) |
| **Certifications** | SOC 2 Type II, ISO 27001 |
| **Server Count** | 50 total (46 active, 4 decommissioned) |

**NOTICE: Lease expires September 2025 - renewal decision required by March 2025**

## 3.1 DC-West Rack Power Density (Selected Racks)

| Rack ID | Row | Alloc kW | Draw kW | Util % | Servers | Primary App |
|---|---|---|---|---|---|---|
| RW-01 | A | 6.0 | 4.1 | 68% | 4 | SupplyChainPro, InventoryMgmt DR |
| RW-02 | A | 6.0 | 3.8 | 63% | 3 | CRM-Enterprise DR standby |
| RW-03 | A | 8.0 | 5.2 | 65% | 4 | ERP-Core DR, FinanceHub DR |
| RW-04 | B | 6.0 | 4.5 | 75% | 4 | Email-Exchange DAG replica |
| RW-05 | B | 6.0 | 3.1 | 52% | 3 | VPN-Gateway, LDAP multi-master |
| RW-06 | B | 8.0 | 6.2 | 78% | 3 | DataWarehouse analytics cluster |
| RW-07 | C | 6.0 | 2.8 | 47% | 3 | Backup and replication targets |

| | | | | | | |
|---|---|---|---|---|---|---|
| RW-08 | C | 6.0 | 4.0 | 67% | 4 | CI-CD-Pipeline build farm |
| RW-09 | C | 6.0 | 3.5 | 58% | 3 | Network appliances (FortiGate, F5) |
| RW-10 | D | 6.0 | 2.2 | 37% | 2 | Spare DR capacity |
| RW-11 | B | 8.0 | 4.1 | 51% | 3 | PayrollEngine DR standby node |
| RW-12 | B | 6.0 | 3.1 | 51% | 1 | TravelBooking DR standby node |
| RW-13 | B | 8.0 | 4.2 | 52% | 2 | E-Commerce-Platform DR standby node |
| RW-14 | B | 6.0 | 3.2 | 53% | 3 | BI-Analytics DR standby node |
| RW-15 | B | 8.0 | 4.4 | 55% | 1 | IT-ServiceDesk DR standby node |
| RW-16 | B | 6.0 | 3.4 | 56% | 2 | LegalDocs DR standby node |
| RW-17 | B | 8.0 | 4.6 | 57% | 3 | API-Gateway DR standby node |
| RW-18 | B | 6.0 | 3.5 | 58% | 1 | ComplianceTracker DR standby node |
| RW-19 | B | 8.0 | 4.7 | 58% | 2 | SecuritySIEM DR standby node |
| RW-20 | B | 6.0 | 3.6 | 60% | 3 | VPN-Gateway DR standby node |
| RW-21 | C | 8.0 | 4.9 | 61% | 1 | DNS-Primary DR standby node |
| RW-22 | C | 6.0 | 3.7 | 61% | 2 | LDAP-Directory DR standby node |
| RW-23 | C | 8.0 | 5.0 | 62% | 3 | FileServer-Central DR standby node |
| RW-24 | C | 6.0 | 3.8 | 63% | 1 | BackupManager DR standby node |
| RW-25 | C | 8.0 | 5.2 | 65% | 2 | MonitoringDashboard DR standby node |
| RW-26 | C | 6.0 | 4.0 | 66% | 3 | SupplyChainPro DR standby node |
| RW-27 | C | 8.0 | 5.4 | 67% | 1 | MarketingAutomation DR standby node |
| RW-28 | C | 6.0 | 4.1 | 68% | 2 | SalesForce-Integration DR standby node |
| RW-29 | C | 8.0 | 5.5 | 68% | 3 | MobileApp-Backend DR standby node |
| RW-30 | C | 6.0 | 4.2 | 70% | 1 | DocumentMgmt DR standby node |
| RW-31 | D | 8.0 | 5.7 | 71% | 2 | VideoConference DR standby node |
| RW-32 | D | 6.0 | 4.3 | 71% | 3 | ProjectTracker DR standby node |
| RW-33 | D | 8.0 | 5.8 | 72% | 1 | QA-TestSuite DR standby node |
| RW-34 | D | 6.0 | 4.4 | 73% | 2 | LogAggregator DR standby node |
| RW-35 | D | 8.0 | 6.0 | 75% | 3 | Microservice-036 DR standby node |
| RW-36 | D | 6.0 | 4.6 | 76% | 1 | Microservice-037 DR standby node |
| RW-37 | D | 8.0 | 6.2 | 77% | 2 | Microservice-038 DR standby node |
| RW-38 | D | 6.0 | 4.7 | 78% | 3 | Microservice-039 DR standby node |
| RW-39 | D | 8.0 | 6.3 | 78% | 1 | Microservice-040 DR standby node |
| RW-40 | D | 6.0 | 4.8 | 80% | 2 | Microservice-041 DR standby node |
| RW-41 | E | 8.0 | 6.5 | 81% | 3 | Microservice-042 DR standby node |
| RW-42 | E | 6.0 | 4.9 | 81% | 1 | Microservice-043 DR standby node |
| RW-43 | E | 8.0 | 6.6 | 82% | 2 | Microservice-044 DR standby node |
| RW-44 | E | 6.0 | 5.0 | 83% | 3 | Microservice-045 DR standby node |
| RW-45 | E | 8.0 | 3.2 | 40% | 1 | Microservice-046 DR standby node |
| RW-46 | E | 6.0 | 2.5 | 41% | 2 | Microservice-047 DR standby node |
| RW-47 | E | 8.0 | 3.4 | 42% | 3 | Microservice-048 DR standby node |
| RW-48 | E | 6.0 | 2.6 | 43% | 1 | Microservice-049 DR standby node |
| RW-49 | E | 8.0 | 3.5 | 43% | 2 | Microservice-050 DR standby node |
| RW-50 | E | 6.0 | 2.7 | 45% | 3 | Microservice-051 DR standby node |
| RW-51 | F | 8.0 | 3.7 | 46% | 1 | Microservice-052 DR standby node |
| RW-52 | F | 6.0 | 2.8 | 46% | 2 | Microservice-053 DR standby node |

# 4. DC-Central - Elk Grove Village, Illinois

The DC-Central facility is located in Elk Grove Village, Illinois, within the O'Hare Technology Park. This is the oldest of the three facilities and serves primarily as a secondary production site and central hub for Midwest regional operations. The aging infrastructure results in a higher PUE compared to the other locations.

| | |
|---|---|
| **Facility Name** | DC-Central |
| **Street Address** | 1200 Estes Avenue, Elk Grove Village, IL 60007 |
| **Colocation Provider** | TierPoint (TP-EGV-1) |
| **Total Floor Space** | 6,000 sq ft (raised floor: 5,400 sq ft) |
| **Total Rack Positions** | 60 racks (42U standard) |
| **Racks in Use** | 48 racks (80% utilized) |
| **Power - Utility Feeds** | Single utility feed from ComEd (redundant via ATS) |
| **Power - Generator Capacity** | 1.5 MW diesel generator (Generac SG150) |
| **Power - Generator Fuel** | 12-hour runtime at full load (2,500 gal) |
| **Power - UPS Capacity** | 1.2 MW UPS (APC Symmetra PX, N+1) |
| **Power - Current Draw** | 0.85 MW (71% of UPS capacity) |
| **Power - PUE** | 1.55 (aging cooling infrastructure) |
| **Power - Cost** | $0.078/kWh (ComEd commercial rate) |
| **Cooling - CRAC Units** | 6 x Liebert CRV 30kW (traditional raised floor) |
| **Cooling - Total Tonnage** | 65 tons refrigeration capacity |
| **Cooling - Containment** | Partial hot aisle containment (rows A-C only) |
| **Cooling - Redundancy** | N+1 (5 required, 6 installed) |
| **Network - Carrier Diversity** | Single carrier: Lumen (dual path within building) |
| **Network - Fiber Paths** | Single fiber path with LTE backup (Verizon) |
| **Network - WAN Bandwidth** | 5 Gbps MPLS + 500 Mbps Internet DIA |
| **Lease - Start Date** | January 1, 2016 |
| **Lease - End Date** | June 30, 2025 |
| **Lease - Renewal Options** | Month-to-month after expiration |
| **Lease - Monthly Cost** | $17,500 (base) + power pass-through |
| **Physical Security** | Badge access, security guard (business hours), CCTV (30-day) |
| **Certifications** | SOC 2 Type II (ISO 27001 lapsed Sep 2024) |
| **Server Count** | 50 total (48 active, 2 decommissioned) |

**CRITICAL: DC-Central lease expires June 30, 2025 - migration planning required**

## 4.1 DC-Central Rack Power Density (Selected Racks)

| Rack ID | Row | Alloc kW | Draw kW | Util % | Servers | Primary App |
|---|---|---|---|---|---|---|
| RC-01 | A | 6.0 | 4.8 | 80% | 4 | LDAP-Directory replica, DNS-Primary |
| RC-02 | A | 6.0 | 5.1 | 85% | 4 | HR-Portal, PayrollEngine backup |
| RC-03 | A | 8.0 | 6.5 | 81% | 4 | DataWarehouse analytics node |
| RC-04 | B | 6.0 | 4.2 | 70% | 3 | IT-ServiceDesk, ComplianceTracker |
| RC-05 | B | 6.0 | 5.5 | 92% | 4 | BI-Analytics, CustomerPortal |
| RC-06 | B | 6.0 | 3.8 | 63% | 3 | FileServer-Central, BackupManager |
| RC-07 | C | 8.0 | 7.0 | 88% | 4 | E-Commerce-Platform secondary |
| RC-08 | C | 6.0 | 4.0 | 67% | 3 | QA-TestSuite, DevOps-Pipeline |
| RC-09 | C | 6.0 | 4.5 | 75% | 4 | Mixed workloads (Tier 3 apps) |

| | | | | | | |
|---|---|---|---|---|---|---|
| RC-10 | D | 6.0 | 3.2 | 53% | 3 | Storage arrays and backup targets |
| RC-11 | B | 8.0 | 3.7 | 46% | 3 | PayrollEngine legacy deployment |
| RC-12 | B | 6.0 | 2.8 | 46% | 1 | TravelBooking legacy deployment |
| RC-13 | B | 8.0 | 3.8 | 47% | 2 | E-Commerce-Platform legacy deployment |
| RC-14 | B | 6.0 | 2.9 | 48% | 3 | BI-Analytics legacy deployment |
| RC-15 | B | 8.0 | 4.0 | 50% | 1 | IT-ServiceDesk legacy deployment |
| RC-16 | B | 6.0 | 3.1 | 51% | 2 | LegalDocs legacy deployment |
| RC-17 | B | 8.0 | 4.2 | 52% | 3 | API-Gateway legacy deployment |
| RC-18 | B | 6.0 | 3.2 | 53% | 1 | ComplianceTracker legacy deployment |
| RC-19 | B | 8.0 | 4.3 | 53% | 2 | SecuritySIEM legacy deployment |
| RC-20 | B | 6.0 | 3.3 | 54% | 3 | VPN-Gateway legacy deployment |
| RC-21 | C | 8.0 | 4.5 | 56% | 1 | DNS-Primary legacy deployment |
| RC-22 | C | 6.0 | 3.4 | 56% | 2 | LDAP-Directory legacy deployment |
| RC-23 | C | 8.0 | 4.6 | 57% | 3 | FileServer-Central legacy deployment |
| RC-24 | C | 6.0 | 3.5 | 58% | 1 | BackupManager legacy deployment |
| RC-25 | C | 8.0 | 4.8 | 60% | 2 | MonitoringDashboard legacy deployment |
| RC-26 | C | 6.0 | 3.7 | 61% | 3 | SupplyChainPro legacy deployment |
| RC-27 | C | 8.0 | 5.0 | 62% | 1 | MarketingAutomation legacy deployment |
| RC-28 | C | 6.0 | 3.8 | 63% | 2 | SalesForce-Integration legacy deployment |
| RC-29 | C | 8.0 | 5.1 | 63% | 3 | MobileApp-Backend legacy deployment |
| RC-30 | C | 6.0 | 3.9 | 65% | 1 | DocumentMgmt legacy deployment |
| RC-31 | D | 8.0 | 5.3 | 66% | 2 | VideoConference legacy deployment |
| RC-32 | D | 6.0 | 4.0 | 66% | 3 | ProjectTracker legacy deployment |
| RC-33 | D | 8.0 | 5.4 | 67% | 1 | QA-TestSuite legacy deployment |
| RC-34 | D | 6.0 | 4.1 | 68% | 2 | LogAggregator legacy deployment |
| RC-35 | D | 8.0 | 5.6 | 70% | 3 | Microservice-036 legacy deployment |
| RC-36 | D | 6.0 | 4.3 | 71% | 1 | Microservice-037 legacy deployment |
| RC-37 | D | 8.0 | 5.8 | 72% | 2 | Microservice-038 legacy deployment |
| RC-38 | D | 6.0 | 4.4 | 73% | 3 | Microservice-039 legacy deployment |
| RC-39 | D | 8.0 | 5.9 | 73% | 1 | Microservice-040 legacy deployment |
| RC-40 | D | 6.0 | 4.5 | 75% | 2 | Microservice-041 legacy deployment |
| RC-41 | E | 8.0 | 6.1 | 76% | 3 | Microservice-042 legacy deployment |
| RC-42 | E | 6.0 | 4.6 | 76% | 1 | Microservice-043 legacy deployment |
| RC-43 | E | 8.0 | 6.2 | 77% | 2 | Microservice-044 legacy deployment |
| RC-44 | E | 6.0 | 4.7 | 78% | 3 | Microservice-045 legacy deployment |
| RC-45 | E | 8.0 | 6.4 | 80% | 1 | Microservice-046 legacy deployment |
| RC-46 | E | 6.0 | 4.9 | 81% | 2 | Microservice-047 legacy deployment |
| RC-47 | E | 8.0 | 6.6 | 82% | 3 | Microservice-048 legacy deployment |
| RC-48 | E | 6.0 | 5.0 | 83% | 1 | Microservice-049 legacy deployment |

# 5. Environmental Monitoring

All three data center facilities are equipped with environmental monitoring sensors (Precision Instruments Corp, VND-028) that continuously track temperature, humidity, and water detection. Alerts are forwarded to PagerDuty and the Datadog monitoring platform.

Temperature Thresholds (per ASHRAE A1 class):
Recommended range: 18-27 C (64-80 F)
Allowable range: 15-32 C (59-90 F)
Warning alert: > 25 C or < 18 C
Critical alert: > 28 C or < 15 C
Emergency shutdown: > 35 C

Humidity Thresholds:
Recommended range: 40-60% RH (dew point: 5.5-15 C)
Warning alert: > 55% RH or < 35% RH
Critical alert: > 65% RH or < 25% RH

Q4 2024 Environmental Summary:
DC-East: Avg temp 21.3 C, 0 temperature alerts, 2 humidity alerts (minor)
DC-West: Avg temp 19.8 C, 0 temperature alerts, 0 humidity alerts
DC-Central: Avg temp 23.5 C, 3 temperature warnings (row D hot spots), 1 humidity alert
Note: DC-Central hot spots in Row D correlate with partial containment coverage

# 6. Physical Security

Physical security measures are implemented in accordance with SOC 2 Trust Service Criteria and ISO 27001 Annex A.11 requirements.

Access Control Layers (all sites):
Layer 1: Perimeter - fence/gate with vehicle barrier (DC-East, DC-West)
Layer 2: Building entry - badge reader + PIN
Layer 3: Cage/suite entry - biometric (fingerprint) + badge
Layer 4: Rack cabinet - individual key locks (select high-security racks)

Surveillance:
DC-East: 32 cameras, 90-day retention, 24/7 NOC monitoring
DC-West: 24 cameras, 60-day retention, 24/7 security guard
DC-Central: 16 cameras, 30-day retention, business hours security guard

Visitor Management:
All visitors require 24-hour advance authorization
Visitors must present government-issued ID
Escort required at all times within the cage/suite area
Visitor logs maintained for 12 months

# 7. Compliance Certifications

| Certification | DC-East | DC-West | DC-Central | Valid Through |
|---|---|---|---|---|
| SOC 2 Type II | Active | Active | Active | Aug 2025 (all sites) |

| | | | | |
|---|---|---|---|---|
| SSAE-18 / SOC 1 | Active | Active | Active | Aug 2025 (all sites) |
| ISO 27001:2022 | Active | Active | Lapsed | Jun 2025 (E/W), Lapsed Sep 2024 (C) |
| PCI-DSS v4.0 | Active | N/A | N/A | Mar 2025 (DC-East only) |
| HIPAA BAA | Active | N/A | N/A | Ongoing (DC-East only) |
| ENERGY STAR | Certified | N/A | N/A | Annual renewal |
| LEED Gold | N/A | Certified | N/A | Permanent certification |

# 8. Capacity Planning Summary

| Metric | DC-East (VA) | DC-West (OR) | DC-Central (IL) | Total |
|---|---|---|---|---|
| Power Capacity (MW) | 2.4 | 1.6 | 1.2 | 5.2 |
| Current Draw (MW) | 1.8 | 0.9 | 0.85 | 3.55 |
| Power Utilization | 75% | 56% | 71% | 68% |
| Total Racks | 120 | 80 | 60 | 260 |
| Racks in Use | 95 | 52 | 48 | 195 |
| Rack Utilization | 79% | 65% | 80% | 75% |
| Floor Space (sq ft) | 12,000 | 8,000 | 6,000 | 26,000 |
| Active Servers | 53 | 46 | 48 | 147 |
| Decommissioned | 2 | 4 | 2 | 8 |
| PUE | 1.45 | 1.35 | 1.55 | 1.45 avg |
| Power Cost ($/kWh) | $0.085 | $0.045 | $0.078 | - |
| Monthly Lease | $35,000 | $18,500 | $17,500 | $71,000 |
| Annual Lease | $420,000 | $222,000 | $210,000 | $852,000 |
| Lease Expiry | Mar 2028 | Sep 2025 | Jun 2025 | - |

Capacity Projections (12-month forecast):

Current total utilization: 68% power, 75% rack space
Projected Q4 2025 utilization (if no consolidation): 72% power, 78% rack space
Projected Q4 2025 utilization (post-consolidation to 2 sites): 82% power, 88% rack space

Consolidation Cost-Benefit Analysis:
  Annual savings from DC-Central closure: ~$280,000 (lease + power + maintenance)
  One-time migration costs: ~$450,000 (included in CAPEX-2025-001)
  Break-even timeline: 19 months post-migration
  Risk: DC-East approaches 90% rack utilization post-consolidation - may require rack optimization

Environmental Considerations:
  DC-East: ENERGY STAR certified, EPA Green Power Partner
  DC-West: 100% renewable energy (hydroelectric), LEED Gold certification
  DC-Central: Partial renewable energy offset (35% wind power credits)
Consolidation to DC-East + DC-West improves overall renewable energy mix from 45% to 62%

# Appendix C: Daily Environmental & PUE Log

The following table provides daily average Power Usage Effectiveness (PUE) and ambient temperature/humidity readings recorded by precision sensors at all three facilities from January 1 through December 31, 2024. This log is used to audit energy efficiency trends.

| Date | East PUE | East Temp | West PUE | West Temp | Cent PUE | Cent Temp |
|---|---|---|---|---|---|---|
| 2024-01-01 | 1.46 | 20.8 C | 1.33 | 20.6 C | 1.54 | 21.4 C |
| 2024-01-02 | 1.46 | 22.3 C | 1.34 | 20.1 C | 1.53 | 22.2 C |
| 2024-01-03 | 1.47 | 21.4 C | 1.36 | 19.1 C | 1.54 | 21.4 C |
| 2024-01-04 | 1.46 | 20.9 C | 1.35 | 19.6 C | 1.57 | 21.8 C |
| 2024-01-05 | 1.45 | 21.7 C | 1.35 | 20.8 C | 1.56 | 22.1 C |
| 2024-01-06 | 1.44 | 22.5 C | 1.35 | 19.0 C | 1.55 | 22.5 C |
| 2024-01-07 | 1.43 | 21.3 C | 1.36 | 20.8 C | 1.57 | 21.3 C |
| 2024-01-08 | 1.44 | 21.5 C | 1.35 | 20.9 C | 1.53 | 22.7 C |
| 2024-01-09 | 1.47 | 21.7 C | 1.35 | 19.6 C | 1.56 | 21.4 C |
| 2024-01-10 | 1.44 | 21.6 C | 1.34 | 19.3 C | 1.54 | 22.4 C |
| 2024-01-11 | 1.46 | 22.1 C | 1.33 | 19.3 C | 1.53 | 21.2 C |
| 2024-01-12 | 1.46 | 21.5 C | 1.33 | 19.3 C | 1.54 | 22.7 C |
| 2024-01-13 | 1.43 | 22.3 C | 1.36 | 19.2 C | 1.56 | 21.1 C |
| 2024-01-14 | 1.43 | 21.2 C | 1.33 | 19.6 C | 1.54 | 22.8 C |
| 2024-01-15 | 1.44 | 22.5 C | 1.35 | 20.9 C | 1.56 | 21.3 C |
| 2024-01-16 | 1.44 | 21.3 C | 1.34 | 21.0 C | 1.54 | 21.0 C |
| 2024-01-17 | 1.47 | 20.9 C | 1.33 | 20.8 C | 1.56 | 22.0 C |
| 2024-01-18 | 1.45 | 21.3 C | 1.35 | 20.0 C | 1.54 | 21.6 C |
| 2024-01-19 | 1.47 | 21.9 C | 1.33 | 20.6 C | 1.56 | 21.4 C |
| 2024-01-20 | 1.44 | 21.1 C | 1.35 | 19.2 C | 1.54 | 21.5 C |
| 2024-01-21 | 1.47 | 21.2 C | 1.35 | 19.0 C | 1.54 | 21.7 C |
| 2024-01-22 | 1.43 | 22.1 C | 1.35 | 20.8 C | 1.54 | 22.8 C |
| 2024-01-23 | 1.47 | 21.4 C | 1.36 | 19.0 C | 1.55 | 21.7 C |
| 2024-01-24 | 1.44 | 21.7 C | 1.34 | 20.0 C | 1.56 | 22.2 C |
| 2024-01-25 | 1.45 | 22.2 C | 1.33 | 19.4 C | 1.55 | 23.0 C |
| 2024-01-26 | 1.44 | 22.0 C | 1.33 | 20.8 C | 1.53 | 22.6 C |
| 2024-01-27 | 1.47 | 20.8 C | 1.35 | 19.7 C | 1.56 | 21.7 C |
| 2024-01-28 | 1.46 | 21.1 C | 1.35 | 20.3 C | 1.56 | 22.8 C |
| 2024-01-29 | 1.45 | 22.4 C | 1.35 | 20.3 C | 1.57 | 22.4 C |
| 2024-01-30 | 1.45 | 20.6 C | 1.35 | 19.7 C | 1.54 | 21.8 C |
| 2024-01-31 | 1.47 | 21.3 C | 1.35 | 19.0 C | 1.54 | 22.5 C |
| 2024-02-01 | 1.47 | 21.5 C | 1.34 | 19.3 C | 1.54 | 22.0 C |
| 2024-02-02 | 1.47 | 21.0 C | 1.36 | 20.1 C | 1.56 | 21.0 C |
| 2024-02-03 | 1.45 | 21.3 C | 1.34 | 19.2 C | 1.55 | 22.1 C |
| 2024-02-04 | 1.46 | 21.7 C | 1.33 | 20.9 C | 1.54 | 21.2 C |
| 2024-02-05 | 1.45 | 22.3 C | 1.33 | 19.9 C | 1.53 | 22.2 C |
| 2024-02-06 | 1.43 | 21.9 C | 1.35 | 20.8 C | 1.55 | 21.4 C |
| 2024-02-07 | 1.46 | 21.9 C | 1.34 | 19.6 C | 1.56 | 21.3 C |
| 2024-02-08 | 1.46 | 20.7 C | 1.34 | 20.7 C | 1.54 | 22.8 C |
| 2024-02-09 | 1.43 | 20.8 C | 1.36 | 20.1 C | 1.56 | 22.1 C |
| 2024-02-10 | 1.44 | 20.6 C | 1.33 | 20.0 C | 1.53 | 21.1 C |
| 2024-02-11 | 1.45 | 22.1 C | 1.34 | 20.0 C | 1.54 | 21.4 C |
| 2024-02-12 | 1.43 | 21.6 C | 1.35 | 19.7 C | 1.56 | 22.8 C |

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|---|---|---|---|---|---|---|
| 2024-02-13 | 1.46 | 20.9 C | 1.34 | 19.5 C | 1.54 | 22.0 C |
| 2024-02-14 | 1.46 | 20.7 C | 1.36 | 20.8 C | 1.57 | 22.5 C |
| 2024-02-15 | 1.44 | 21.7 C | 1.33 | 19.4 C | 1.57 | 21.7 C |
| 2024-02-16 | 1.44 | 22.4 C | 1.36 | 19.5 C | 1.56 | 21.2 C |
| 2024-02-17 | 1.47 | 21.5 C | 1.35 | 20.7 C | 1.54 | 21.5 C |
| 2024-02-18 | 1.45 | 22.2 C | 1.34 | 19.7 C | 1.54 | 22.8 C |
| 2024-02-19 | 1.45 | 22.0 C | 1.34 | 19.5 C | 1.54 | 21.2 C |
| 2024-02-20 | 1.46 | 21.9 C | 1.34 | 21.0 C | 1.55 | 22.5 C |
| 2024-02-21 | 1.47 | 21.5 C | 1.34 | 19.0 C | 1.56 | 23.0 C |
| 2024-02-22 | 1.44 | 21.1 C | 1.33 | 19.0 C | 1.55 | 22.7 C |
| 2024-02-23 | 1.44 | 22.2 C | 1.33 | 19.2 C | 1.54 | 22.1 C |
| 2024-02-24 | 1.46 | 20.9 C | 1.35 | 19.6 C | 1.56 | 22.1 C |
| 2024-02-25 | 1.45 | 20.8 C | 1.36 | 20.0 C | 1.53 | 22.2 C |
| 2024-02-26 | 1.43 | 20.6 C | 1.34 | 19.1 C | 1.56 | 21.9 C |
| 2024-02-27 | 1.47 | 21.3 C | 1.35 | 19.1 C | 1.57 | 21.8 C |
| 2024-02-28 | 1.46 | 21.6 C | 1.34 | 20.8 C | 1.55 | 21.9 C |
| 2024-02-29 | 1.45 | 20.9 C | 1.35 | 19.1 C | 1.56 | 21.6 C |
| 2024-03-01 | 1.45 | 21.4 C | 1.33 | 20.8 C | 1.56 | 21.6 C |
| 2024-03-02 | 1.47 | 21.3 C | 1.34 | 19.6 C | 1.55 | 21.5 C |
| 2024-03-03 | 1.44 | 20.7 C | 1.36 | 19.3 C | 1.56 | 21.9 C |
| 2024-03-04 | 1.47 | 21.4 C | 1.35 | 20.6 C | 1.54 | 21.7 C |
| 2024-03-05 | 1.43 | 22.2 C | 1.35 | 19.3 C | 1.54 | 21.9 C |
| 2024-03-06 | 1.47 | 21.6 C | 1.36 | 19.1 C | 1.53 | 21.3 C |
| 2024-03-07 | 1.43 | 22.2 C | 1.36 | 20.0 C | 1.55 | 22.1 C |
| 2024-03-08 | 1.45 | 21.0 C | 1.36 | 19.6 C | 1.53 | 22.1 C |
| 2024-03-09 | 1.46 | 21.2 C | 1.34 | 19.2 C | 1.55 | 21.3 C |
| 2024-03-10 | 1.44 | 22.0 C | 1.35 | 20.5 C | 1.54 | 22.0 C |
| 2024-03-11 | 1.46 | 21.9 C | 1.35 | 19.8 C | 1.55 | 21.1 C |
| 2024-03-12 | 1.43 | 20.7 C | 1.35 | 19.3 C | 1.56 | 21.4 C |
| 2024-03-13 | 1.44 | 21.4 C | 1.34 | 20.7 C | 1.55 | 21.7 C |
| 2024-03-14 | 1.46 | 21.7 C | 1.34 | 19.2 C | 1.54 | 22.4 C |
| 2024-03-15 | 1.46 | 22.1 C | 1.36 | 20.1 C | 1.56 | 22.9 C |
| 2024-03-16 | 1.44 | 20.9 C | 1.33 | 20.3 C | 1.55 | 21.2 C |
| 2024-03-17 | 1.45 | 20.9 C | 1.35 | 20.6 C | 1.54 | 21.4 C |
| 2024-03-18 | 1.45 | 21.7 C | 1.35 | 19.5 C | 1.57 | 22.4 C |
| 2024-03-19 | 1.44 | 22.4 C | 1.33 | 20.8 C | 1.57 | 22.6 C |
| 2024-03-20 | 1.44 | 20.9 C | 1.34 | 20.9 C | 1.53 | 22.0 C |
| 2024-03-21 | 1.44 | 22.2 C | 1.33 | 20.0 C | 1.54 | 22.3 C |
| 2024-03-22 | 1.44 | 20.7 C | 1.33 | 20.5 C | 1.57 | 22.0 C |
| 2024-03-23 | 1.47 | 21.6 C | 1.34 | 20.2 C | 1.56 | 22.8 C |
| 2024-03-24 | 1.46 | 22.1 C | 1.34 | 20.4 C | 1.55 | 22.7 C |
| 2024-03-25 | 1.44 | 22.2 C | 1.33 | 20.0 C | 1.54 | 21.3 C |
| 2024-03-26 | 1.43 | 20.7 C | 1.36 | 19.8 C | 1.56 | 21.6 C |
| 2024-03-27 | 1.45 | 20.7 C | 1.35 | 20.6 C | 1.55 | 21.5 C |
| 2024-03-28 | 1.47 | 22.2 C | 1.35 | 20.0 C | 1.53 | 21.1 C |
| 2024-03-29 | 1.46 | 21.2 C | 1.35 | 19.9 C | 1.55 | 21.9 C |
| 2024-03-30 | 1.45 | 20.5 C | 1.36 | 19.4 C | 1.54 | 21.4 C |
| 2024-03-31 | 1.44 | 20.9 C | 1.36 | 20.7 C | 1.57 | 22.3 C |
| 2024-04-01 | 1.44 | 21.3 C | 1.36 | 20.3 C | 1.57 | 21.3 C |
| 2024-04-02 | 1.44 | 21.7 C | 1.33 | 20.4 C | 1.54 | 21.1 C |
| 2024-04-03 | 1.46 | 22.2 C | 1.35 | 19.4 C | 1.54 | 21.2 C |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2024-04-04 | 1.46 | 22.1 C | 1.36 | 20.2 C | 1.54 | 22.7 C |
| 2024-04-05 | 1.44 | 21.9 C | 1.35 | 19.4 C | 1.53 | 21.6 C |
| 2024-04-06 | 1.46 | 22.2 C | 1.33 | 20.6 C | 1.54 | 21.3 C |
| 2024-04-07 | 1.44 | 21.4 C | 1.36 | 20.4 C | 1.54 | 21.5 C |
| 2024-04-08 | 1.44 | 22.5 C | 1.36 | 19.3 C | 1.53 | 21.8 C |
| 2024-04-09 | 1.45 | 22.4 C | 1.35 | 19.2 C | 1.55 | 22.0 C |
| 2024-04-10 | 1.45 | 21.5 C | 1.36 | 19.2 C | 1.56 | 22.1 C |
| 2024-04-11 | 1.45 | 20.5 C | 1.35 | 19.1 C | 1.56 | 21.9 C |
| 2024-04-12 | 1.46 | 22.2 C | 1.34 | 19.0 C | 1.54 | 22.7 C |
| 2024-04-13 | 1.47 | 22.3 C | 1.34 | 19.9 C | 1.55 | 22.2 C |
| 2024-04-14 | 1.46 | 22.1 C | 1.34 | 20.6 C | 1.55 | 21.5 C |
| 2024-04-15 | 1.44 | 21.1 C | 1.34 | 20.5 C | 1.56 | 21.4 C |
| 2024-04-16 | 1.45 | 20.9 C | 1.33 | 19.3 C | 1.57 | 23.0 C |
| 2024-04-17 | 1.46 | 21.1 C | 1.36 | 20.3 C | 1.53 | 22.7 C |
| 2024-04-18 | 1.46 | 22.1 C | 1.34 | 19.3 C | 1.56 | 22.5 C |
| 2024-04-19 | 1.45 | 22.3 C | 1.34 | 20.3 C | 1.56 | 21.7 C |
| 2024-04-20 | 1.45 | 21.0 C | 1.35 | 19.5 C | 1.56 | 22.8 C |
| 2024-04-21 | 1.45 | 22.2 C | 1.35 | 19.9 C | 1.56 | 21.5 C |
| 2024-04-22 | 1.45 | 21.6 C | 1.33 | 19.4 C | 1.56 | 21.6 C |
| 2024-04-23 | 1.43 | 21.3 C | 1.34 | 21.0 C | 1.54 | 22.5 C |
| 2024-04-24 | 1.45 | 21.8 C | 1.34 | 20.5 C | 1.56 | 22.8 C |
| 2024-04-25 | 1.44 | 21.0 C | 1.35 | 20.2 C | 1.57 | 21.3 C |
| 2024-04-26 | 1.45 | 21.3 C | 1.34 | 20.3 C | 1.55 | 21.2 C |
| 2024-04-27 | 1.45 | 21.7 C | 1.34 | 19.4 C | 1.56 | 22.0 C |
| 2024-04-28 | 1.45 | 21.7 C | 1.35 | 20.0 C | 1.54 | 22.5 C |
| 2024-04-29 | 1.45 | 20.7 C | 1.35 | 20.8 C | 1.54 | 21.2 C |
| 2024-04-30 | 1.46 | 21.9 C | 1.34 | 20.6 C | 1.57 | 21.7 C |
| 2024-05-01 | 1.44 | 22.3 C | 1.34 | 20.7 C | 1.53 | 21.0 C |
| 2024-05-02 | 1.44 | 22.2 C | 1.34 | 20.2 C | 1.54 | 21.6 C |
| 2024-05-03 | 1.44 | 22.4 C | 1.35 | 20.1 C | 1.56 | 21.7 C |
| 2024-05-04 | 1.43 | 21.5 C | 1.35 | 20.4 C | 1.56 | 21.9 C |
| 2024-05-05 | 1.47 | 21.1 C | 1.35 | 20.8 C | 1.56 | 22.8 C |
| 2024-05-06 | 1.44 | 21.7 C | 1.34 | 19.4 C | 1.54 | 21.4 C |
| 2024-05-07 | 1.45 | 21.2 C | 1.36 | 20.6 C | 1.55 | 22.7 C |
| 2024-05-08 | 1.43 | 20.7 C | 1.34 | 19.5 C | 1.56 | 21.6 C |
| 2024-05-09 | 1.46 | 21.5 C | 1.36 | 20.7 C | 1.54 | 22.6 C |
| 2024-05-10 | 1.45 | 22.0 C | 1.35 | 20.9 C | 1.56 | 21.2 C |
| 2024-05-11 | 1.44 | 22.0 C | 1.33 | 19.8 C | 1.54 | 22.9 C |
| 2024-05-12 | 1.44 | 21.7 C | 1.33 | 20.6 C | 1.54 | 21.5 C |
| 2024-05-13 | 1.44 | 22.4 C | 1.36 | 20.0 C | 1.56 | 21.4 C |
| 2024-05-14 | 1.46 | 22.3 C | 1.33 | 19.1 C | 1.54 | 22.4 C |
| 2024-05-15 | 1.43 | 21.1 C | 1.35 | 20.3 C | 1.55 | 22.8 C |
| 2024-05-16 | 1.44 | 20.9 C | 1.34 | 19.0 C | 1.54 | 22.0 C |
| 2024-05-17 | 1.43 | 22.0 C | 1.33 | 19.6 C | 1.56 | 21.7 C |
| 2024-05-18 | 1.46 | 21.4 C | 1.34 | 19.9 C | 1.56 | 21.6 C |
| 2024-05-19 | 1.44 | 20.5 C | 1.35 | 19.0 C | 1.54 | 22.3 C |
| 2024-05-20 | 1.45 | 21.1 C | 1.34 | 20.7 C | 1.57 | 22.1 C |
| 2024-05-21 | 1.44 | 21.4 C | 1.35 | 20.2 C | 1.55 | 22.9 C |
| 2024-05-22 | 1.46 | 22.2 C | 1.35 | 19.5 C | 1.54 | 21.5 C |
| 2024-05-23 | 1.44 | 21.7 C | 1.33 | 19.8 C | 1.56 | 22.6 C |
| 2024-05-24 | 1.46 | 20.9 C | 1.34 | 20.1 C | 1.57 | 21.7 C |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2024-05-25 | 1.43 | 22.1 C | 1.34 | 19.3 C | 1.53 | 21.0 C |
| 2024-05-26 | 1.44 | 22.3 C | 1.34 | 19.5 C | 1.55 | 21.2 C |
| 2024-05-27 | 1.46 | 22.1 C | 1.34 | 19.5 C | 1.54 | 22.4 C |
| 2024-05-28 | 1.44 | 20.6 C | 1.36 | 20.8 C | 1.56 | 21.3 C |
| 2024-05-29 | 1.46 | 20.6 C | 1.33 | 20.7 C | 1.55 | 22.5 C |
| 2024-05-30 | 1.44 | 20.7 C | 1.36 | 19.6 C | 1.54 | 22.2 C |
| 2024-05-31 | 1.46 | 20.9 C | 1.34 | 20.8 C | 1.55 | 21.7 C |
| 2024-06-01 | 1.43 | 22.1 C | 1.34 | 19.7 C | 1.56 | 21.5 C |
| 2024-06-02 | 1.46 | 21.4 C | 1.35 | 19.7 C | 1.56 | 21.4 C |
| 2024-06-03 | 1.44 | 22.0 C | 1.35 | 19.2 C | 1.57 | 22.3 C |
| 2024-06-04 | 1.45 | 21.3 C | 1.35 | 19.5 C | 1.54 | 22.6 C |
| 2024-06-05 | 1.45 | 21.3 C | 1.34 | 20.9 C | 1.56 | 21.4 C |
| 2024-06-06 | 1.44 | 20.9 C | 1.34 | 20.5 C | 1.54 | 22.8 C |
| 2024-06-07 | 1.44 | 22.2 C | 1.34 | 20.3 C | 1.57 | 21.6 C |
| 2024-06-08 | 1.45 | 21.3 C | 1.36 | 20.0 C | 1.55 | 21.5 C |
| 2024-06-09 | 1.44 | 21.2 C | 1.33 | 20.7 C | 1.54 | 21.1 C |
| 2024-06-10 | 1.44 | 21.0 C | 1.36 | 20.7 C | 1.57 | 22.8 C |
| 2024-06-11 | 1.46 | 21.0 C | 1.35 | 20.0 C | 1.57 | 23.0 C |
| 2024-06-12 | 1.43 | 21.2 C | 1.34 | 19.9 C | 1.57 | 22.5 C |
| 2024-06-13 | 1.43 | 21.8 C | 1.34 | 20.2 C | 1.56 | 22.0 C |
| 2024-06-14 | 1.46 | 21.1 C | 1.35 | 20.4 C | 1.54 | 22.9 C |
| 2024-06-15 | 1.47 | 22.2 C | 1.34 | 19.6 C | 1.56 | 21.9 C |
| 2024-06-16 | 1.44 | 21.2 C | 1.33 | 20.6 C | 1.55 | 21.7 C |
| 2024-06-17 | 1.47 | 20.7 C | 1.35 | 19.2 C | 1.57 | 22.7 C |
| 2024-06-18 | 1.44 | 21.6 C | 1.33 | 20.4 C | 1.56 | 22.1 C |
| 2024-06-19 | 1.46 | 21.2 C | 1.35 | 19.8 C | 1.56 | 21.4 C |
| 2024-06-20 | 1.46 | 20.8 C | 1.35 | 19.4 C | 1.56 | 21.0 C |
| 2024-06-21 | 1.43 | 21.2 C | 1.35 | 19.6 C | 1.55 | 22.4 C |
| 2024-06-22 | 1.44 | 21.4 C | 1.34 | 21.0 C | 1.54 | 22.5 C |
| 2024-06-23 | 1.44 | 20.7 C | 1.35 | 19.2 C | 1.56 | 21.3 C |
| 2024-06-24 | 1.47 | 20.9 C | 1.35 | 19.1 C | 1.55 | 22.7 C |
| 2024-06-25 | 1.47 | 22.2 C | 1.35 | 19.9 C | 1.55 | 22.9 C |
| 2024-06-26 | 1.43 | 21.7 C | 1.34 | 19.5 C | 1.54 | 21.7 C |
| 2024-06-27 | 1.44 | 21.9 C | 1.35 | 20.7 C | 1.55 | 21.9 C |
| 2024-06-28 | 1.46 | 21.8 C | 1.35 | 19.3 C | 1.56 | 21.8 C |
| 2024-06-29 | 1.47 | 21.1 C | 1.36 | 19.8 C | 1.54 | 22.2 C |
| 2024-06-30 | 1.45 | 22.5 C | 1.35 | 19.8 C | 1.54 | 22.2 C |
| 2024-07-01 | 1.46 | 21.8 C | 1.35 | 20.8 C | 1.55 | 21.6 C |
| 2024-07-02 | 1.47 | 21.1 C | 1.35 | 20.0 C | 1.56 | 22.0 C |
| 2024-07-03 | 1.45 | 21.5 C | 1.36 | 19.9 C | 1.55 | 21.3 C |
| 2024-07-04 | 1.47 | 20.9 C | 1.34 | 20.2 C | 1.56 | 21.0 C |
| 2024-07-05 | 1.46 | 22.2 C | 1.35 | 19.1 C | 1.53 | 21.9 C |
| 2024-07-06 | 1.47 | 22.0 C | 1.35 | 19.5 C | 1.55 | 21.4 C |
| 2024-07-07 | 1.45 | 22.0 C | 1.36 | 20.0 C | 1.57 | 22.2 C |
| 2024-07-08 | 1.45 | 22.4 C | 1.36 | 19.3 C | 1.57 | 22.7 C |
| 2024-07-09 | 1.47 | 22.1 C | 1.35 | 19.3 C | 1.55 | 21.5 C |
| 2024-07-10 | 1.46 | 22.3 C | 1.33 | 20.2 C | 1.55 | 22.7 C |
| 2024-07-11 | 1.44 | 22.0 C | 1.36 | 20.7 C | 1.55 | 22.1 C |
| 2024-07-12 | 1.44 | 20.7 C | 1.34 | 19.7 C | 1.53 | 22.5 C |
| 2024-07-13 | 1.44 | 21.3 C | 1.36 | 20.4 C | 1.53 | 21.8 C |
| 2024-07-14 | 1.44 | 21.3 C | 1.35 | 19.9 C | 1.55 | 22.9 C |

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| 2024-07-15 | 1.46 | 21.1 C | 1.35 | 20.3 C | 1.55 | 21.5 C |
| 2024-07-16 | 1.44 | 21.5 C | 1.33 | 20.5 C | 1.56 | 22.4 C |
| 2024-07-17 | 1.45 | 22.4 C | 1.35 | 20.2 C | 1.55 | 21.2 C |
| 2024-07-18 | 1.46 | 21.2 C | 1.35 | 19.7 C | 1.55 | 22.2 C |
| 2024-07-19 | 1.46 | 21.4 C | 1.35 | 21.0 C | 1.55 | 21.8 C |
| 2024-07-20 | 1.46 | 21.1 C | 1.34 | 19.1 C | 1.57 | 21.3 C |
| 2024-07-21 | 1.43 | 20.9 C | 1.34 | 20.1 C | 1.54 | 22.1 C |
| 2024-07-22 | 1.46 | 20.5 C | 1.35 | 19.3 C | 1.57 | 22.5 C |
| 2024-07-23 | 1.44 | 21.8 C | 1.34 | 20.5 C | 1.54 | 21.9 C |
| 2024-07-24 | 1.45 | 20.6 C | 1.34 | 19.3 C | 1.55 | 21.5 C |
| 2024-07-25 | 1.47 | 22.4 C | 1.35 | 19.2 C | 1.55 | 22.7 C |
| 2024-07-26 | 1.44 | 21.2 C | 1.35 | 20.4 C | 1.54 | 22.4 C |
| 2024-07-27 | 1.45 | 21.0 C | 1.34 | 19.4 C | 1.56 | 22.7 C |
| 2024-07-28 | 1.44 | 22.4 C | 1.35 | 19.8 C | 1.57 | 22.9 C |
| 2024-07-29 | 1.47 | 21.4 C | 1.35 | 20.8 C | 1.54 | 21.9 C |
| 2024-07-30 | 1.43 | 21.5 C | 1.35 | 20.9 C | 1.55 | 22.3 C |
| 2024-07-31 | 1.44 | 22.4 C | 1.35 | 20.0 C | 1.54 | 22.6 C |
| 2024-08-01 | 1.46 | 21.8 C | 1.33 | 19.7 C | 1.54 | 21.4 C |
| 2024-08-02 | 1.45 | 20.7 C | 1.36 | 19.7 C | 1.54 | 23.0 C |
| 2024-08-03 | 1.45 | 21.7 C | 1.34 | 20.3 C | 1.54 | 21.2 C |
| 2024-08-04 | 1.46 | 21.2 C | 1.35 | 20.2 C | 1.54 | 22.8 C |
| 2024-08-05 | 1.44 | 20.6 C | 1.35 | 20.7 C | 1.55 | 22.5 C |
| 2024-08-06 | 1.44 | 21.1 C | 1.35 | 20.0 C | 1.55 | 21.1 C |
| 2024-08-07 | 1.47 | 22.2 C | 1.35 | 19.2 C | 1.55 | 22.0 C |
| 2024-08-08 | 1.45 | 22.3 C | 1.33 | 19.8 C | 1.54 | 21.2 C |
| 2024-08-09 | 1.44 | 20.7 C | 1.34 | 20.7 C | 1.54 | 22.6 C |
| 2024-08-10 | 1.46 | 21.1 C | 1.34 | 20.3 C | 1.56 | 21.5 C |
| 2024-08-11 | 1.44 | 21.0 C | 1.35 | 19.9 C | 1.55 | 21.5 C |
| 2024-08-12 | 1.46 | 21.9 C | 1.33 | 19.7 C | 1.53 | 21.5 C |
| 2024-08-13 | 1.44 | 21.5 C | 1.35 | 19.2 C | 1.56 | 21.2 C |
| 2024-08-14 | 1.44 | 21.2 C | 1.35 | 20.6 C | 1.54 | 22.7 C |
| 2024-08-15 | 1.46 | 21.0 C | 1.35 | 19.8 C | 1.57 | 21.1 C |
| 2024-08-16 | 1.43 | 21.3 C | 1.36 | 19.2 C | 1.56 | 22.4 C |
| 2024-08-17 | 1.43 | 22.1 C | 1.34 | 20.0 C | 1.54 | 21.5 C |
| 2024-08-18 | 1.47 | 21.0 C | 1.35 | 19.2 C | 1.55 | 22.9 C |
| 2024-08-19 | 1.47 | 21.3 C | 1.34 | 20.9 C | 1.56 | 21.1 C |
| 2024-08-20 | 1.44 | 22.0 C | 1.34 | 19.9 C | 1.54 | 21.6 C |
| 2024-08-21 | 1.43 | 21.5 C | 1.33 | 20.0 C | 1.56 | 21.9 C |
| 2024-08-22 | 1.45 | 21.5 C | 1.34 | 19.3 C | 1.53 | 22.4 C |
| 2024-08-23 | 1.46 | 21.9 C | 1.36 | 20.1 C | 1.54 | 22.8 C |
| 2024-08-24 | 1.46 | 22.3 C | 1.35 | 20.3 C | 1.54 | 22.4 C |
| 2024-08-25 | 1.46 | 21.4 C | 1.35 | 19.9 C | 1.55 | 22.6 C |
| 2024-08-26 | 1.45 | 20.7 C | 1.34 | 19.2 C | 1.56 | 21.3 C |
| 2024-08-27 | 1.43 | 21.4 C | 1.33 | 19.3 C | 1.56 | 22.2 C |
| 2024-08-28 | 1.46 | 22.4 C | 1.33 | 19.6 C | 1.54 | 22.0 C |
| 2024-08-29 | 1.47 | 20.9 C | 1.36 | 19.5 C | 1.54 | 21.3 C |
| 2024-08-30 | 1.44 | 22.5 C | 1.36 | 20.6 C | 1.54 | 21.5 C |
| 2024-08-31 | 1.47 | 22.1 C | 1.35 | 19.4 C | 1.56 | 22.4 C |
| 2024-09-01 | 1.45 | 21.2 C | 1.34 | 20.7 C | 1.56 | 22.0 C |
| 2024-09-02 | 1.44 | 21.1 C | 1.33 | 19.5 C | 1.56 | 22.1 C |
| 2024-09-03 | 1.45 | 21.7 C | 1.36 | 19.2 C | 1.54 | 21.7 C |

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| 2024-09-04 | 1.46 | 21.0 C | 1.36 | 19.6 C | 1.55 | 22.2 C |
| 2024-09-05 | 1.46 | 22.4 C | 1.36 | 20.4 C | 1.53 | 21.0 C |
| 2024-09-06 | 1.45 | 21.1 C | 1.33 | 19.8 C | 1.56 | 21.8 C |
| 2024-09-07 | 1.46 | 22.0 C | 1.36 | 20.1 C | 1.54 | 21.1 C |
| 2024-09-08 | 1.44 | 22.1 C | 1.34 | 20.1 C | 1.53 | 22.2 C |
| 2024-09-09 | 1.44 | 21.2 C | 1.35 | 20.6 C | 1.57 | 21.7 C |
| 2024-09-10 | 1.44 | 20.8 C | 1.34 | 19.8 C | 1.53 | 22.1 C |
| 2024-09-11 | 1.44 | 21.5 C | 1.34 | 19.8 C | 1.56 | 22.8 C |
| 2024-09-12 | 1.47 | 21.6 C | 1.35 | 19.3 C | 1.55 | 21.1 C |
| 2024-09-13 | 1.45 | 21.1 C | 1.34 | 19.7 C | 1.53 | 21.0 C |
| 2024-09-14 | 1.44 | 22.2 C | 1.36 | 20.3 C | 1.57 | 21.6 C |
| 2024-09-15 | 1.46 | 21.0 C | 1.34 | 19.5 C | 1.54 | 21.1 C |
| 2024-09-16 | 1.45 | 22.3 C | 1.35 | 20.4 C | 1.55 | 22.2 C |
| 2024-09-17 | 1.45 | 21.1 C | 1.34 | 20.1 C | 1.57 | 22.6 C |
| 2024-09-18 | 1.46 | 21.4 C | 1.35 | 19.3 C | 1.55 | 22.2 C |
| 2024-09-19 | 1.46 | 21.2 C | 1.34 | 20.9 C | 1.54 | 21.0 C |
| 2024-09-20 | 1.46 | 22.2 C | 1.34 | 20.5 C | 1.55 | 22.9 C |
| 2024-09-21 | 1.46 | 20.6 C | 1.36 | 20.6 C | 1.55 | 22.8 C |
| 2024-09-22 | 1.45 | 20.8 C | 1.35 | 19.6 C | 1.54 | 22.4 C |
| 2024-09-23 | 1.45 | 21.3 C | 1.35 | 20.4 C | 1.57 | 21.1 C |
| 2024-09-24 | 1.47 | 22.4 C | 1.36 | 20.3 C | 1.56 | 21.2 C |
| 2024-09-25 | 1.46 | 22.5 C | 1.34 | 20.0 C | 1.55 | 21.8 C |
| 2024-09-26 | 1.47 | 21.7 C | 1.33 | 20.1 C | 1.53 | 22.2 C |
| 2024-09-27 | 1.46 | 21.0 C | 1.34 | 20.5 C | 1.53 | 21.5 C |
| 2024-09-28 | 1.45 | 20.6 C | 1.36 | 19.7 C | 1.56 | 21.4 C |
| 2024-09-29 | 1.47 | 21.7 C | 1.34 | 19.7 C | 1.53 | 22.7 C |
| 2024-09-30 | 1.45 | 21.9 C | 1.35 | 19.6 C | 1.55 | 21.8 C |
| 2024-10-01 | 1.45 | 21.5 C | 1.33 | 19.9 C | 1.54 | 22.3 C |
| 2024-10-02 | 1.44 | 20.7 C | 1.35 | 19.8 C | 1.56 | 21.8 C |
| 2024-10-03 | 1.47 | 21.1 C | 1.34 | 20.7 C | 1.54 | 22.5 C |
| 2024-10-04 | 1.46 | 20.8 C | 1.35 | 21.0 C | 1.54 | 21.4 C |
| 2024-10-05 | 1.44 | 21.5 C | 1.34 | 19.9 C | 1.54 | 21.1 C |
| 2024-10-06 | 1.44 | 22.1 C | 1.35 | 19.7 C | 1.54 | 21.0 C |
| 2024-10-07 | 1.43 | 21.0 C | 1.35 | 20.0 C | 1.53 | 22.1 C |
| 2024-10-08 | 1.44 | 21.3 C | 1.34 | 19.3 C | 1.56 | 21.4 C |
| 2024-10-09 | 1.44 | 21.4 C | 1.33 | 20.6 C | 1.55 | 21.1 C |
| 2024-10-10 | 1.46 | 21.2 C | 1.35 | 20.6 C | 1.55 | 23.0 C |
| 2024-10-11 | 1.47 | 22.0 C | 1.35 | 19.9 C | 1.53 | 22.1 C |
| 2024-10-12 | 1.44 | 22.2 C | 1.34 | 19.4 C | 1.56 | 22.9 C |
| 2024-10-13 | 1.46 | 20.7 C | 1.36 | 19.8 C | 1.55 | 22.0 C |
| 2024-10-14 | 1.43 | 21.1 C | 1.34 | 19.8 C | 1.56 | 21.4 C |
| 2024-10-15 | 1.45 | 20.7 C | 1.34 | 19.2 C | 1.56 | 22.5 C |
| 2024-10-16 | 1.45 | 21.5 C | 1.35 | 19.8 C | 1.57 | 21.6 C |
| 2024-10-17 | 1.44 | 21.3 C | 1.35 | 20.3 C | 1.56 | 21.7 C |
| 2024-10-18 | 1.46 | 20.6 C | 1.34 | 20.8 C | 1.55 | 21.9 C |
| 2024-10-19 | 1.44 | 20.6 C | 1.36 | 20.2 C | 1.54 | 22.0 C |
| 2024-10-20 | 1.44 | 20.9 C | 1.36 | 20.0 C | 1.54 | 21.2 C |
| 2024-10-21 | 1.45 | 20.9 C | 1.34 | 20.7 C | 1.55 | 21.2 C |
| 2024-10-22 | 1.47 | 22.5 C | 1.34 | 21.0 C | 1.54 | 21.1 C |
| 2024-10-23 | 1.47 | 22.1 C | 1.35 | 19.7 C | 1.54 | 22.8 C |
| 2024-10-24 | 1.46 | 22.1 C | 1.33 | 20.2 C | 1.56 | 21.2 C |

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|---|---|---|---|---|---|---|
| 2024-10-25 | 1.45 | 22.2 C | 1.34 | 19.3 C | 1.56 | 21.2 C |
| 2024-10-26 | 1.45 | 22.0 C | 1.36 | 20.8 C | 1.53 | 21.8 C |
| 2024-10-27 | 1.44 | 22.3 C | 1.34 | 20.1 C | 1.56 | 21.9 C |
| 2024-10-28 | 1.44 | 22.2 C | 1.35 | 19.7 C | 1.57 | 21.5 C |
| 2024-10-29 | 1.45 | 21.7 C | 1.34 | 20.0 C | 1.55 | 22.6 C |
| 2024-10-30 | 1.47 | 20.8 C | 1.34 | 21.0 C | 1.55 | 21.9 C |
| 2024-10-31 | 1.46 | 21.5 C | 1.33 | 20.6 C | 1.55 | 21.1 C |
| 2024-11-01 | 1.45 | 21.1 C | 1.35 | 19.3 C | 1.55 | 21.2 C |
| 2024-11-02 | 1.46 | 20.6 C | 1.34 | 19.1 C | 1.54 | 22.4 C |
| 2024-11-03 | 1.45 | 22.4 C | 1.34 | 20.5 C | 1.53 | 22.4 C |
| 2024-11-04 | 1.44 | 21.8 C | 1.34 | 20.7 C | 1.57 | 21.6 C |
| 2024-11-05 | 1.43 | 21.0 C | 1.35 | 19.5 C | 1.54 | 22.7 C |
| 2024-11-06 | 1.45 | 22.1 C | 1.35 | 20.8 C | 1.55 | 21.2 C |
| 2024-11-07 | 1.45 | 21.2 C | 1.34 | 19.2 C | 1.55 | 21.7 C |
| 2024-11-08 | 1.46 | 21.5 C | 1.35 | 20.7 C | 1.54 | 22.3 C |
| 2024-11-09 | 1.45 | 22.4 C | 1.35 | 21.0 C | 1.55 | 21.1 C |
| 2024-11-10 | 1.45 | 22.3 C | 1.34 | 20.2 C | 1.54 | 21.7 C |
| 2024-11-11 | 1.47 | 21.8 C | 1.35 | 19.9 C | 1.56 | 21.6 C |
| 2024-11-12 | 1.45 | 21.2 C | 1.34 | 20.1 C | 1.56 | 21.5 C |
| 2024-11-13 | 1.44 | 21.6 C | 1.35 | 19.5 C | 1.56 | 21.3 C |
| 2024-11-14 | 1.47 | 22.1 C | 1.36 | 19.2 C | 1.55 | 22.1 C |
| 2024-11-15 | 1.44 | 22.0 C | 1.35 | 20.0 C | 1.57 | 21.9 C |
| 2024-11-16 | 1.44 | 21.8 C | 1.34 | 20.8 C | 1.56 | 21.8 C |
| 2024-11-17 | 1.45 | 22.1 C | 1.34 | 20.4 C | 1.55 | 22.9 C |
| 2024-11-18 | 1.46 | 21.0 C | 1.36 | 19.0 C | 1.56 | 23.0 C |
| 2024-11-19 | 1.44 | 20.6 C | 1.35 | 19.8 C | 1.56 | 21.0 C |
| 2024-11-20 | 1.43 | 20.6 C | 1.34 | 20.2 C | 1.55 | 22.1 C |
| 2024-11-21 | 1.45 | 21.6 C | 1.35 | 20.7 C | 1.55 | 21.5 C |
| 2024-11-22 | 1.44 | 21.4 C | 1.35 | 21.0 C | 1.54 | 22.3 C |
| 2024-11-23 | 1.45 | 22.2 C | 1.36 | 19.3 C | 1.56 | 22.7 C |
| 2024-11-24 | 1.44 | 20.7 C | 1.34 | 19.6 C | 1.57 | 21.6 C |
| 2024-11-25 | 1.44 | 21.7 C | 1.35 | 20.5 C | 1.55 | 22.0 C |
| 2024-11-26 | 1.47 | 21.1 C | 1.34 | 20.8 C | 1.53 | 21.6 C |
| 2024-11-27 | 1.44 | 20.8 C | 1.36 | 19.2 C | 1.54 | 21.1 C |
| 2024-11-28 | 1.43 | 21.8 C | 1.33 | 19.1 C | 1.54 | 22.8 C |
| 2024-11-29 | 1.47 | 20.6 C | 1.34 | 19.1 C | 1.54 | 21.1 C |
| 2024-11-30 | 1.46 | 21.6 C | 1.34 | 20.2 C | 1.55 | 21.5 C |
| 2024-12-01 | 1.47 | 20.6 C | 1.35 | 19.3 C | 1.56 | 21.7 C |
| 2024-12-02 | 1.43 | 20.6 C | 1.34 | 19.1 C | 1.57 | 21.4 C |
| 2024-12-03 | 1.46 | 21.3 C | 1.34 | 20.2 C | 1.53 | 22.9 C |
| 2024-12-04 | 1.43 | 21.6 C | 1.35 | 19.4 C | 1.53 | 22.2 C |
| 2024-12-05 | 1.46 | 21.9 C | 1.34 | 19.8 C | 1.57 | 21.7 C |
| 2024-12-06 | 1.44 | 21.0 C | 1.35 | 19.5 C | 1.54 | 22.2 C |
| 2024-12-07 | 1.45 | 22.0 C | 1.35 | 19.7 C | 1.57 | 22.9 C |
| 2024-12-08 | 1.44 | 20.8 C | 1.35 | 20.0 C | 1.56 | 22.9 C |
| 2024-12-09 | 1.45 | 20.5 C | 1.35 | 20.9 C | 1.55 | 22.6 C |
| 2024-12-10 | 1.47 | 21.2 C | 1.33 | 19.5 C | 1.55 | 22.6 C |
| 2024-12-11 | 1.45 | 21.4 C | 1.34 | 20.0 C | 1.56 | 22.7 C |
| 2024-12-12 | 1.45 | 22.1 C | 1.36 | 19.2 C | 1.54 | 22.1 C |
| 2024-12-13 | 1.46 | 21.0 C | 1.35 | 19.2 C | 1.56 | 22.5 C |
| 2024-12-14 | 1.44 | 21.2 C | 1.36 | 19.5 C | 1.54 | 21.2 C |

# Appendix D: Detailed Power Distribution Unit Inventory

This appendix provides a detailed inventory of all Power Distribution Units (PDUs) installed across the three datacenter facilities. Each PDU is mapped to its associated rack, circuit breaker panel, and UPS feed. The inventory was last validated during the October 2024 annual facility audit.

PDU naming convention: PDU-[DC]-[Row][Rack]-[A/B] where A/B denotes primary/redundant power feed. All production racks are equipped with dual PDUs for N+1 redundancy. Development and staging racks may operate with single PDU configurations.

| PDU ID | DC Location | Rack | Feed | Capacity kW | Load kW | Util % | UPS Feed | Circuit Panel |
|---|---|---|---|---|---|---|---|---|
| PDU-E-R01-A | DC-East Virgin | RE-01 | A | 25.0 | 9.7 | 38.8 | UPS-E-1 | PNL-E-01 |
| PDU-E-R01-B | DC-East Virgin | RE-01 | B | 20.0 | 7.3 | 36.5 | UPS-E-1 | PNL-E-01 |
| PDU-E-R02-A | DC-East Virgin | RE-02 | A | 25.0 | 9.8 | 39.2 | UPS-E-1 | PNL-E-01 |
| PDU-E-R02-B | DC-East Virgin | RE-02 | B | 10.0 | 6.5 | 65.0 | UPS-E-1 | PNL-E-01 |
| PDU-E-R03-A | DC-East Virgin | RE-03 | A | 25.0 | 15.3 | 61.2 | UPS-E-1 | PNL-E-02 |
| PDU-E-R03-B | DC-East Virgin | RE-03 | B | 15.0 | 5.6 | 37.3 | UPS-E-1 | PNL-E-02 |
| PDU-E-R04-A | DC-East Virgin | RE-04 | A | 25.0 | 18.6 | 74.4 | UPS-E-1 | PNL-E-02 |
| PDU-E-R04-B | DC-East Virgin | RE-04 | B | 20.0 | 7.5 | 37.5 | UPS-E-1 | PNL-E-02 |
| PDU-E-R05-A | DC-East Virgin | RE-05 | A | 10.0 | 5.3 | 53.0 | UPS-E-2 | PNL-E-02 |
| PDU-E-R05-B | DC-East Virgin | RE-05 | B | 10.0 | 8.4 | 84.0 | UPS-E-2 | PNL-E-02 |
| PDU-E-R06-A | DC-East Virgin | RE-06 | A | 15.0 | 8.4 | 56.0 | UPS-E-2 | PNL-E-03 |
| PDU-E-R06-B | DC-East Virgin | RE-06 | B | 20.0 | 16.9 | 84.5 | UPS-E-2 | PNL-E-03 |
| PDU-E-R07-A | DC-East Virgin | RE-07 | A | 20.0 | 11.3 | 56.5 | UPS-E-2 | PNL-E-03 |
| PDU-E-R07-B | DC-East Virgin | RE-07 | B | 15.0 | 11.7 | 78.0 | UPS-E-2 | PNL-E-03 |
| PDU-E-R08-A | DC-East Virgin | RE-08 | A | 25.0 | 19.3 | 77.2 | UPS-E-2 | PNL-E-03 |
| PDU-E-R08-B | DC-East Virgin | RE-08 | B | 25.0 | 13.3 | 53.2 | UPS-E-2 | PNL-E-03 |
| PDU-E-R09-A | DC-East Virgin | RE-09 | A | 15.0 | 8.6 | 57.3 | UPS-E-2 | PNL-E-04 |
| PDU-E-R09-B | DC-East Virgin | RE-09 | B | 25.0 | 14.1 | 56.4 | UPS-E-2 | PNL-E-04 |
| PDU-E-R10-A | DC-East Virgin | RE-10 | A | 15.0 | 12.3 | 82.0 | UPS-E-3 | PNL-E-04 |
| PDU-E-R10-B | DC-East Virgin | RE-10 | B | 15.0 | 11.6 | 77.3 | UPS-E-3 | PNL-E-04 |
| PDU-E-R11-A | DC-East Virgin | RE-11 | A | 15.0 | 9.6 | 64.0 | UPS-E-3 | PNL-E-04 |
| PDU-E-R11-B | DC-East Virgin | RE-11 | B | 25.0 | 16.7 | 66.8 | UPS-E-3 | PNL-E-04 |
| PDU-E-R12-A | DC-East Virgin | RE-12 | A | 15.0 | 11.6 | 77.3 | UPS-E-3 | PNL-E-05 |
| PDU-E-R12-B | DC-East Virgin | RE-12 | B | 15.0 | 6.5 | 43.3 | UPS-E-3 | PNL-E-05 |
| PDU-E-R13-A | DC-East Virgin | RE-13 | A | 20.0 | 8.5 | 42.5 | UPS-E-3 | PNL-E-05 |
| PDU-E-R13-B | DC-East Virgin | RE-13 | B | 20.0 | 10.3 | 51.5 | UPS-E-3 | PNL-E-05 |
| PDU-E-R14-A | DC-East Virgin | RE-14 | A | 15.0 | 12.3 | 82.0 | UPS-E-3 | PNL-E-05 |
| PDU-E-R14-B | DC-East Virgin | RE-14 | B | 25.0 | 9.0 | 36.0 | UPS-E-3 | PNL-E-05 |
| PDU-E-R15-A | DC-East Virgin | RE-15 | A | 20.0 | 7.3 | 36.5 | UPS-E-4 | PNL-E-06 |
| PDU-E-R15-B | DC-East Virgin | RE-15 | B | 25.0 | 12.3 | 49.2 | UPS-E-4 | PNL-E-06 |
| PDU-E-R16-A | DC-East Virgin | RE-16 | A | 15.0 | 10.9 | 72.7 | UPS-E-4 | PNL-E-06 |
| PDU-E-R16-B | DC-East Virgin | RE-16 | B | 10.0 | 4.5 | 45.0 | UPS-E-4 | PNL-E-06 |
| PDU-E-R17-A | DC-East Virgin | RE-17 | A | 20.0 | 13.8 | 69.0 | UPS-E-4 | PNL-E-06 |
| PDU-E-R17-B | DC-East Virgin | RE-17 | B | 20.0 | 11.6 | 58.0 | UPS-E-4 | PNL-E-06 |
| PDU-E-R18-A | DC-East Virgin | RE-18 | A | 15.0 | 11.7 | 78.0 | UPS-E-4 | PNL-E-07 |
| PDU-E-R18-B | DC-East Virgin | RE-18 | B | 25.0 | 20.2 | 80.8 | UPS-E-4 | PNL-E-07 |
| PDU-E-R19-A | DC-East Virgin | RE-19 | A | 25.0 | 17.4 | 69.6 | UPS-E-4 | PNL-E-07 |
| PDU-E-R19-B | DC-East Virgin | RE-19 | B | 15.0 | 11.6 | 77.3 | UPS-E-4 | PNL-E-07 |
| PDU-E-R20-A | DC-East Virgin | RE-20 | A | 25.0 | 17.1 | 68.4 | UPS-E-5 | PNL-E-07 |
| PDU-E-R20-B | DC-East Virgin | RE-20 | B | 15.0 | 6.8 | 45.3 | UPS-E-5 | PNL-E-07 |
| PDU-E-R21-A | DC-East Virgin | RE-21 | A | 15.0 | 6.4 | 42.7 | UPS-E-5 | PNL-E-08 |
| PDU-E-R21-B | DC-East Virgin | RE-21 | B | 20.0 | 16.8 | 84.0 | UPS-E-5 | PNL-E-08 |
| PDU-E-R22-A | DC-East Virgin | RE-22 | A | 15.0 | 9.5 | 63.3 | UPS-E-5 | PNL-E-08 |
| PDU-E-R22-B | DC-East Virgin | RE-22 | B | 25.0 | 9.8 | 39.2 | UPS-E-5 | PNL-E-08 |
| PDU-E-R23-A | DC-East Virgin | RE-23 | A | 25.0 | 12.7 | 50.8 | UPS-E-5 | PNL-E-08 |
| PDU-E-R23-B | DC-East Virgin | RE-23 | B | 25.0 | 20.8 | 83.2 | UPS-E-5 | PNL-E-08 |
| PDU-E-R24-A | DC-East Virgin | RE-24 | A | 25.0 | 13.9 | 55.6 | UPS-E-5 | PNL-E-09 |
| PDU-E-R24-B | DC-East Virgin | RE-24 | B | 20.0 | 8.6 | 43.0 | UPS-E-5 | PNL-E-09 |
| PDU-E-R25-A | DC-East Virgin | RE-25 | A | 20.0 | 16.0 | 80.0 | UPS-E-6 | PNL-E-09 |
| PDU-E-R25-B | DC-East Virgin | RE-25 | B | 15.0 | 11.8 | 78.7 | UPS-E-6 | PNL-E-09 |
| PDU-W-R01-A | DC-West Oregon | RW-01 | A | 20.0 | 16.6 | 83.0 | UPS-W-1 | PNL-W-01 |
| PDU-W-R01-B | DC-West Oregon | RW-01 | B | 10.0 | 8.3 | 83.0 | UPS-W-1 | PNL-W-01 |
| PDU-W-R02-A | DC-West Oregon | RW-02 | A | 20.0 | 10.0 | 50.0 | UPS-W-1 | PNL-W-01 |
| PDU-W-R02-B | DC-West Oregon | RW-02 | B | 25.0 | 10.0 | 40.0 | UPS-W-1 | PNL-W-01 |

| PDU ID | DC Location | Rack | Feed | Capacity kW | Load kW | Util % | UPS Feed | Circuit Panel |
|---|---|---|---|---|---|---|---|---|
| PDU-W-R03-A | DC-West Oregon | RW-03 | A | 15.0 | 7.6 | 50.7 | UPS-W-1 | PNL-W-02 |
| PDU-W-R03-B | DC-West Oregon | RW-03 | B | 15.0 | 7.2 | 48.0 | UPS-W-1 | PNL-W-02 |
| PDU-W-R04-A | DC-West Oregon | RW-04 | A | 10.0 | 6.6 | 66.0 | UPS-W-1 | PNL-W-02 |
| PDU-W-R04-B | DC-West Oregon | RW-04 | B | 25.0 | 10.8 | 43.2 | UPS-W-1 | PNL-W-02 |
| PDU-W-R05-A | DC-West Oregon | RW-05 | A | 20.0 | 14.9 | 74.5 | UPS-W-2 | PNL-W-02 |
| PDU-W-R05-B | DC-West Oregon | RW-05 | B | 20.0 | 16.5 | 82.5 | UPS-W-2 | PNL-W-02 |
| PDU-W-R06-A | DC-West Oregon | RW-06 | A | 20.0 | 16.2 | 81.0 | UPS-W-2 | PNL-W-03 |
| PDU-W-R06-B | DC-West Oregon | RW-06 | B | 15.0 | 12.7 | 84.7 | UPS-W-2 | PNL-W-03 |
| PDU-W-R07-A | DC-West Oregon | RW-07 | A | 20.0 | 15.3 | 76.5 | UPS-W-2 | PNL-W-03 |
| PDU-W-R07-B | DC-West Oregon | RW-07 | B | 15.0 | 7.3 | 48.7 | UPS-W-2 | PNL-W-03 |
| PDU-W-R08-A | DC-West Oregon | RW-08 | A | 15.0 | 8.0 | 53.3 | UPS-W-2 | PNL-W-03 |
| PDU-W-R08-B | DC-West Oregon | RW-08 | B | 10.0 | 5.0 | 50.0 | UPS-W-2 | PNL-W-03 |
| PDU-W-R09-A | DC-West Oregon | RW-09 | A | 20.0 | 16.5 | 82.5 | UPS-W-2 | PNL-W-04 |
| PDU-W-R09-B | DC-West Oregon | RW-09 | B | 20.0 | 12.8 | 64.0 | UPS-W-2 | PNL-W-04 |
| PDU-W-R10-A | DC-West Oregon | RW-10 | A | 10.0 | 5.4 | 54.0 | UPS-W-3 | PNL-W-04 |
| PDU-W-R10-B | DC-West Oregon | RW-10 | B | 15.0 | 6.9 | 46.0 | UPS-W-3 | PNL-W-04 |
| PDU-W-R11-A | DC-West Oregon | RW-11 | A | 20.0 | 16.8 | 84.0 | UPS-W-3 | PNL-W-04 |
| PDU-W-R11-B | DC-West Oregon | RW-11 | B | 10.0 | 6.8 | 68.0 | UPS-W-3 | PNL-W-04 |
| PDU-W-R12-A | DC-West Oregon | RW-12 | A | 20.0 | 7.2 | 36.0 | UPS-W-3 | PNL-W-05 |
| PDU-W-R12-B | DC-West Oregon | RW-12 | B | 10.0 | 6.1 | 61.0 | UPS-W-3 | PNL-W-05 |
| PDU-W-R13-A | DC-West Oregon | RW-13 | A | 25.0 | 16.4 | 65.6 | UPS-W-3 | PNL-W-05 |
| PDU-W-R13-B | DC-West Oregon | RW-13 | B | 20.0 | 9.5 | 47.5 | UPS-W-3 | PNL-W-05 |
| PDU-W-R14-A | DC-West Oregon | RW-14 | A | 20.0 | 11.6 | 58.0 | UPS-W-3 | PNL-W-05 |
| PDU-W-R14-B | DC-West Oregon | RW-14 | B | 15.0 | 9.4 | 62.7 | UPS-W-3 | PNL-W-05 |
| PDU-W-R15-A | DC-West Oregon | RW-15 | A | 20.0 | 8.7 | 43.5 | UPS-W-4 | PNL-W-06 |
| PDU-W-R15-B | DC-West Oregon | RW-15 | B | 15.0 | 7.4 | 49.3 | UPS-W-4 | PNL-W-06 |
| PDU-W-R16-A | DC-West Oregon | RW-16 | A | 10.0 | 6.9 | 69.0 | UPS-W-4 | PNL-W-06 |
| PDU-W-R16-B | DC-West Oregon | RW-16 | B | 20.0 | 11.6 | 58.0 | UPS-W-4 | PNL-W-06 |
| PDU-W-R17-A | DC-West Oregon | RW-17 | A | 25.0 | 16.4 | 65.6 | UPS-W-4 | PNL-W-06 |
| PDU-W-R17-B | DC-West Oregon | RW-17 | B | 20.0 | 11.8 | 59.0 | UPS-W-4 | PNL-W-06 |
| PDU-W-R18-A | DC-West Oregon | RW-18 | A | 25.0 | 12.2 | 48.8 | UPS-W-4 | PNL-W-07 |
| PDU-W-R18-B | DC-West Oregon | RW-18 | B | 25.0 | 14.7 | 58.8 | UPS-W-4 | PNL-W-07 |
| PDU-W-R19-A | DC-West Oregon | RW-19 | A | 10.0 | 7.5 | 75.0 | UPS-W-4 | PNL-W-07 |
| PDU-W-R19-B | DC-West Oregon | RW-19 | B | 20.0 | 12.6 | 63.0 | UPS-W-4 | PNL-W-07 |
| PDU-W-R20-A | DC-West Oregon | RW-20 | A | 15.0 | 12.2 | 81.3 | UPS-W-5 | PNL-W-07 |
| PDU-W-R20-B | DC-West Oregon | RW-20 | B | 25.0 | 16.0 | 64.0 | UPS-W-5 | PNL-W-07 |
| PDU-C-R01-A | DC-Central III | RC-01 | A | 15.0 | 9.1 | 60.7 | UPS-C-1 | PNL-C-01 |
| PDU-C-R01-B | DC-Central III | RC-01 | B | 25.0 | 10.7 | 42.8 | UPS-C-1 | PNL-C-01 |
| PDU-C-R02-A | DC-Central III | RC-02 | A | 20.0 | 11.0 | 55.0 | UPS-C-1 | PNL-C-01 |
| PDU-C-R02-B | DC-Central III | RC-02 | B | 25.0 | 13.0 | 52.0 | UPS-C-1 | PNL-C-01 |
| PDU-C-R03-A | DC-Central III | RC-03 | A | 10.0 | 5.3 | 53.0 | UPS-C-1 | PNL-C-02 |
| PDU-C-R03-B | DC-Central III | RC-03 | B | 20.0 | 9.3 | 46.5 | UPS-C-1 | PNL-C-02 |
| PDU-C-R04-A | DC-Central III | RC-04 | A | 15.0 | 6.0 | 40.0 | UPS-C-1 | PNL-C-02 |
| PDU-C-R04-B | DC-Central III | RC-04 | B | 25.0 | 18.6 | 74.4 | UPS-C-1 | PNL-C-02 |
| PDU-C-R05-A | DC-Central III | RC-05 | A | 10.0 | 4.4 | 44.0 | UPS-C-2 | PNL-C-02 |
| PDU-C-R05-B | DC-Central III | RC-05 | B | 15.0 | 7.8 | 52.0 | UPS-C-2 | PNL-C-02 |
| PDU-C-R06-A | DC-Central III | RC-06 | A | 15.0 | 5.9 | 39.3 | UPS-C-2 | PNL-C-03 |
| PDU-C-R06-B | DC-Central III | RC-06 | B | 15.0 | 6.9 | 46.0 | UPS-C-2 | PNL-C-03 |
| PDU-C-R07-A | DC-Central III | RC-07 | A | 10.0 | 3.9 | 39.0 | UPS-C-2 | PNL-C-03 |
| PDU-C-R07-B | DC-Central III | RC-07 | B | 20.0 | 14.5 | 72.5 | UPS-C-2 | PNL-C-03 |
| PDU-C-R08-A | DC-Central III | RC-08 | A | 15.0 | 8.8 | 58.7 | UPS-C-2 | PNL-C-03 |
| PDU-C-R08-B | DC-Central III | RC-08 | B | 20.0 | 13.5 | 67.5 | UPS-C-2 | PNL-C-03 |
| PDU-C-R09-A | DC-Central III | RC-09 | A | 20.0 | 7.0 | 35.0 | UPS-C-2 | PNL-C-04 |
| PDU-C-R09-B | DC-Central III | RC-09 | B | 25.0 | 10.7 | 42.8 | UPS-C-2 | PNL-C-04 |
| PDU-C-R10-A | DC-Central III | RC-10 | A | 10.0 | 7.5 | 75.0 | UPS-C-3 | PNL-C-04 |
| PDU-C-R10-B | DC-Central III | RC-10 | B | 20.0 | 7.5 | 37.5 | UPS-C-3 | PNL-C-04 |
| PDU-C-R11-A | DC-Central III | RC-11 | A | 20.0 | 16.5 | 82.5 | UPS-C-3 | PNL-C-04 |
| PDU-C-R11-B | DC-Central III | RC-11 | B | 10.0 | 7.6 | 76.0 | UPS-C-3 | PNL-C-04 |
| PDU-C-R12-A | DC-Central III | RC-12 | A | 10.0 | 7.2 | 72.0 | UPS-C-3 | PNL-C-05 |
| PDU-C-R12-B | DC-Central III | RC-12 | B | 25.0 | 9.1 | 36.4 | UPS-C-3 | PNL-C-05 |
| PDU-C-R13-A | DC-Central III | RC-13 | A | 20.0 | 11.2 | 56.0 | UPS-C-3 | PNL-C-05 |
| PDU-C-R13-B | DC-Central III | RC-13 | B | 20.0 | 7.5 | 37.5 | UPS-C-3 | PNL-C-05 |
| PDU-C-R14-A | DC-Central III | RC-14 | A | 20.0 | 10.6 | 53.0 | UPS-C-3 | PNL-C-05 |
| PDU-C-R14-B | DC-Central III | RC-14 | B | 25.0 | 13.6 | 54.4 | UPS-C-3 | PNL-C-05 |

| PDU ID | DC Location | Rack | Feed | Capacity kW | Load kW | Util % | UPS Feed | Circuit Panel |
|---|---|---|---|---|---|---|---|---|
| PDU-C-R15-A | DC-Central III | RC-15 | A | 15.0 | 7.4 | 49.3 | UPS-C-4 | PNL-C-06 |
| PDU-C-R15-B | DC-Central III | RC-15 | B | 10.0 | 4.1 | 41.0 | UPS-C-4 | PNL-C-06 |

# Appendix E: HVAC System Configuration and Maintenance Log

This appendix documents the heating, ventilation, and air conditioning (HVAC) systems at all three datacenter facilities. Each CRAC/CRAH unit is documented with its configuration parameters, maintenance schedule, and Q4 2024 maintenance completion status.

The cooling architecture at DC-East and DC-West utilizes chilled water CRAH units with hot aisle containment. DC-Central Illinois employs a mixed strategy with direct expansion (DX) CRAC units in the legacy wing and chilled water CRAH units in the newer expansion.

| Unit ID | DC Location | Type | Capacity kW | Set Point C | Actual C | Maint Due | Maint Status |
|---|---|---|---|---|---|---|---|
| HVAC-E-01 | DC-East Virginia | CRAH | 75 | 20.4 | 20.7 | 2025-02-14 | Scheduled |
| HVAC-E-02 | DC-East Virginia | CRAH | 30 | 20.7 | 21.6 | 2025-02-09 | Scheduled |
| HVAC-E-03 | DC-East Virginia | CRAH | 75 | 21.3 | 22.0 | 2025-02-19 | Scheduled |
| HVAC-E-04 | DC-East Virginia | CRAH | 100 | 20.8 | 22.4 | 2024-12-09 | Completed |
| HVAC-E-05 | DC-East Virginia | CRAH | 30 | 21.2 | 20.1 | 2025-03-17 | Scheduled |
| HVAC-E-06 | DC-East Virginia | CRAH | 100 | 21.6 | 20.6 | 2024-12-18 | Completed |
| HVAC-E-07 | DC-East Virginia | CRAH | 50 | 20.6 | 19.7 | 2024-10-12 | Completed |
| HVAC-E-08 | DC-East Virginia | CRAH | 75 | 20.6 | 22.3 | 2025-02-02 | Scheduled |
| HVAC-E-09 | DC-East Virginia | CRAH | 100 | 20.3 | 20.2 | 2025-02-14 | Scheduled |
| HVAC-E-10 | DC-East Virginia | CRAH | 75 | 21.0 | 21.2 | 2025-01-10 | Scheduled |
| HVAC-E-11 | DC-East Virginia | CRAH | 100 | 20.7 | 22.8 | 2024-10-14 | Completed |
| HVAC-E-12 | DC-East Virginia | CRAH | 30 | 20.5 | 19.9 | 2024-12-05 | Completed |
| HVAC-E-13 | DC-East Virginia | CRAH | 100 | 20.6 | 21.7 | 2025-01-07 | Scheduled |
| HVAC-E-14 | DC-East Virginia | CRAH | 50 | 21.0 | 22.6 | 2025-03-09 | Scheduled |
| HVAC-E-15 | DC-East Virginia | CRAH | 50 | 21.1 | 22.1 | 2025-03-21 | Scheduled |
| HVAC-E-16 | DC-East Virginia | CRAH | 30 | 21.7 | 20.9 | 2024-11-18 | Completed |
| HVAC-E-17 | DC-East Virginia | CRAH | 100 | 20.6 | 20.4 | 2024-10-24 | Completed |
| HVAC-E-18 | DC-East Virginia | CRAH | 100 | 20.1 | 20.2 | 2024-11-06 | Completed |
| HVAC-W-01 | DC-West Oregon | CRAH | 30 | 21.8 | 22.9 | 2025-02-26 | Scheduled |
| HVAC-W-02 | DC-West Oregon | CRAH | 30 | 21.7 | 21.0 | 2025-02-12 | Scheduled |
| HVAC-W-03 | DC-West Oregon | CRAH | 100 | 21.5 | 22.4 | 2024-10-14 | Completed |
| HVAC-W-04 | DC-West Oregon | CRAH | 100 | 20.3 | 22.6 | 2024-11-12 | Completed |
| HVAC-W-05 | DC-West Oregon | CRAH | 50 | 21.0 | 22.0 | 2025-03-26 | Scheduled |
| HVAC-W-06 | DC-West Oregon | CRAH | 75 | 22.0 | 22.3 | 2024-11-11 | Completed |
| HVAC-W-07 | DC-West Oregon | CRAH | 50 | 20.3 | 20.2 | 2025-03-14 | Scheduled |
| HVAC-W-08 | DC-West Oregon | CRAH | 75 | 21.3 | 20.3 | 2025-01-08 | Scheduled |
| HVAC-W-09 | DC-West Oregon | CRAH | 50 | 21.5 | 20.5 | 2024-10-25 | Completed |
| HVAC-W-10 | DC-West Oregon | CRAH | 50 | 20.7 | 21.2 | 2025-03-06 | Scheduled |
| HVAC-W-11 | DC-West Oregon | CRAH | 30 | 21.1 | 22.0 | 2024-12-21 | Scheduled |
| HVAC-W-12 | DC-West Oregon | CRAH | 75 | 22.0 | 20.6 | 2024-11-28 | Completed |
| HVAC-W-13 | DC-West Oregon | CRAH | 100 | 20.4 | 19.4 | 2024-10-14 | Completed |
| HVAC-W-14 | DC-West Oregon | CRAH | 75 | 21.4 | 23.5 | 2025-02-28 | Scheduled |
| HVAC-C-01 | DC-Central Illin | CRAC | 75 | 20.3 | 22.2 | 2025-01-01 | Scheduled |
| HVAC-C-02 | DC-Central Illin | CRAC | 50 | 20.2 | 21.7 | 2024-12-24 | Scheduled |
| HVAC-C-03 | DC-Central Illin | CRAC | 75 | 22.0 | 23.9 | 2025-01-25 | Scheduled |
| HVAC-C-04 | DC-Central Illin | CRAC | 100 | 20.1 | 21.2 | 2024-10-05 | Completed |
| HVAC-C-05 | DC-Central Illin | CRAC | 75 | 21.9 | 21.1 | 2025-02-13 | Scheduled |
| HVAC-C-06 | DC-Central Illin | CRAH | 50 | 21.1 | 20.2 | 2025-03-14 | Scheduled |
| HVAC-C-07 | DC-Central Illin | CRAH | 30 | 21.0 | 21.1 | 2025-03-09 | Scheduled |
| HVAC-C-08 | DC-Central Illin | CRAH | 30 | 21.7 | 23.8 | 2024-12-01 | Completed |
| HVAC-C-09 | DC-Central Illin | CRAH | 100 | 21.8 | 23.5 | 2024-10-26 | Completed |
| HVAC-C-10 | DC-Central Illin | CRAH | 50 | 20.9 | 22.0 | 2025-01-22 | Scheduled |

# Appendix F: Network Cabling Plant Documentation

This appendix provides the structured cabling documentation for all three datacenter facilities. The cabling plant includes fiber optic backbone (single-mode and multi-mode), Category 6A copper cabling for management networks, and out-of-band serial console cabling. All cabling was tested and certified per TIA-568.2 standards.

Cable management is organized by distribution zones. Each zone is served by an Intermediate Distribution Frame (IDF) connected to the Main Distribution Frame (MDF) via redundant fiber paths. The cable database is maintained in the DCIM platform and was last reconciled with physical plant during the Q3 2024 audit.

| Cable ID | DC Location | Type | From | To | Length m | Test Result | Install Date |
|---|---|---|---|---|---|---|---|
| CBL-E-0001 | DC-East Virgin | Serial Conso | MDF-E | RE-02/P5 | 43 | Pass | 2021-07-23 |
| CBL-E-0002 | DC-East Virgin | MM Fiber OM4 | MDF-E | RE-03/P2 | 13 | Pass | 2018-08-21 |
| CBL-E-0003 | DC-East Virgin | Cat 6A STP | MDF-E | RE-04/P42 | 34 | Pass | 2022-10-18 |
| CBL-E-0004 | DC-East Virgin | Cat 6A STP | MDF-E | RE-05/P7 | 53 | Pass | 2019-06-29 |
| CBL-E-0005 | DC-East Virgin | Cat 6A UTP | MDF-E | RE-06/P20 | 36 | Pass | 2023-02-19 |
| CBL-E-0006 | DC-East Virgin | MM Fiber OM4 | IDF-E-01 | RE-07/P40 | 27 | Pass | 2023-11-28 |
| CBL-E-0007 | DC-East Virgin | SM Fiber | IDF-E-01 | RE-08/P5 | 17 | Pass | 2020-12-13 |
| CBL-E-0008 | DC-East Virgin | SM Fiber | IDF-E-01 | RE-09/P46 | 54 | Pass | 2019-08-12 |
| CBL-E-0009 | DC-East Virgin | Serial Conso | IDF-E-01 | RE-10/P32 | 63 | Pass | 2021-09-29 |
| CBL-E-0010 | DC-East Virgin | Cat 6A UTP | IDF-E-02 | RE-11/P18 | 24 | Pass | 2018-01-02 |
| CBL-E-0011 | DC-East Virgin | MM Fiber OM4 | IDF-E-02 | RE-12/P13 | 7 | Pass | 2019-12-01 |
| CBL-E-0012 | DC-East Virgin | SM Fiber | IDF-E-02 | RE-13/P11 | 13 | Pass | 2020-09-16 |
| CBL-E-0013 | DC-East Virgin | Cat 6A STP | IDF-E-02 | RE-14/P12 | 13 | Pass | 2018-02-13 |
| CBL-E-0014 | DC-East Virgin | Cat 6A UTP | IDF-E-02 | RE-15/P14 | 51 | Pass | 2019-05-06 |
| CBL-E-0015 | DC-East Virgin | Serial Conso | IDF-E-02 | RE-16/P10 | 46 | Pass | 2024-06-04 |
| CBL-E-0016 | DC-East Virgin | MM Fiber OM4 | IDF-E-02 | RE-17/P32 | 49 | Pass | 2023-11-11 |
| CBL-E-0017 | DC-East Virgin | SM Fiber | IDF-E-02 | RE-18/P2 | 76 | Pass | 2021-09-25 |
| CBL-E-0018 | DC-East Virgin | Serial Conso | IDF-E-02 | RE-19/P6 | 80 | Pass | 2020-08-26 |
| CBL-E-0019 | DC-East Virgin | Cat 6A UTP | IDF-E-02 | RE-20/P1 | 57 | Pass | 2020-08-19 |
| CBL-E-0020 | DC-East Virgin | MM Fiber OM4 | IDF-E-03 | RE-21/P27 | 73 | Pass | 2018-12-23 |
| CBL-E-0021 | DC-East Virgin | Serial Conso | IDF-E-03 | RE-22/P43 | 46 | Pass | 2019-08-27 |
| CBL-E-0022 | DC-East Virgin | SM Fiber | IDF-E-03 | RE-23/P24 | 46 | Pass | 2020-07-08 |
| CBL-E-0023 | DC-East Virgin | SM Fiber | IDF-E-03 | RE-24/P41 | 66 | Pass | 2023-03-02 |
| CBL-E-0024 | DC-East Virgin | MM Fiber OM4 | IDF-E-03 | RE-25/P5 | 69 | Pass | 2023-03-13 |
| CBL-E-0025 | DC-East Virgin | SM Fiber | IDF-E-03 | RE-01/P30 | 5 | Pass | 2022-10-09 |
| CBL-E-0026 | DC-East Virgin | Cat 6A STP | IDF-E-03 | RE-02/P38 | 22 | Pass | 2024-04-09 |
| CBL-E-0027 | DC-East Virgin | MM Fiber OM4 | IDF-E-03 | RE-03/P18 | 23 | Pass | 2023-08-31 |
| CBL-E-0028 | DC-East Virgin | Cat 6A UTP | IDF-E-03 | RE-04/P19 | 56 | Pass | 2022-05-14 |
| CBL-E-0029 | DC-East Virgin | Cat 6A STP | IDF-E-03 | RE-05/P35 | 11 | Pass | 2023-09-05 |
| CBL-E-0030 | DC-East Virgin | MM Fiber OM4 | IDF-E-04 | RE-06/P12 | 34 | Pass | 2020-05-25 |
| CBL-E-0031 | DC-East Virgin | Cat 6A STP | IDF-E-04 | RE-07/P32 | 17 | Pass | 2023-04-21 |
| CBL-E-0032 | DC-East Virgin | SM Fiber | IDF-E-04 | RE-08/P31 | 38 | Pass | 2021-11-27 |
| CBL-E-0033 | DC-East Virgin | Cat 6A UTP | IDF-E-04 | RE-09/P1 | 34 | Pass | 2019-10-25 |
| CBL-E-0034 | DC-East Virgin | Serial Conso | IDF-E-04 | RE-10/P2 | 40 | Pass | 2019-04-11 |
| CBL-E-0035 | DC-East Virgin | Cat 6A STP | IDF-E-04 | RE-11/P34 | 78 | Pass | 2018-11-11 |
| CBL-E-0036 | DC-East Virgin | MM Fiber OM4 | IDF-E-04 | RE-12/P9 | 85 | Pass | 2019-08-18 |
| CBL-E-0037 | DC-East Virgin | Serial Conso | IDF-E-04 | RE-13/P35 | 85 | Pass | 2023-08-16 |
| CBL-E-0038 | DC-East Virgin | MM Fiber OM4 | IDF-E-04 | RE-14/P45 | 32 | Pass | 2019-12-22 |
| CBL-E-0039 | DC-East Virgin | Cat 6A UTP | IDF-E-04 | RE-15/P7 | 11 | Pass | 2019-09-25 |
| CBL-E-0040 | DC-East Virgin | Serial Conso | IDF-E-05 | RE-16/P14 | 4 | Pass | 2019-01-15 |
| CBL-E-0041 | DC-East Virgin | Serial Conso | IDF-E-05 | RE-17/P6 | 38 | Pass | 2018-04-01 |
| CBL-E-0042 | DC-East Virgin | MM Fiber OM4 | IDF-E-05 | RE-18/P18 | 44 | Pass | 2023-10-31 |
| CBL-E-0043 | DC-East Virgin | Serial Conso | IDF-E-05 | RE-19/P10 | 79 | Pass | 2018-10-07 |
| CBL-E-0044 | DC-East Virgin | Cat 6A UTP | IDF-E-05 | RE-20/P35 | 35 | Pass | 2023-12-05 |
| CBL-E-0045 | DC-East Virgin | Cat 6A UTP | IDF-E-05 | RE-21/P8 | 13 | Pass | 2021-10-04 |
| CBL-E-0046 | DC-East Virgin | SM Fiber | IDF-E-05 | RE-22/P3 | 70 | Pass | 2019-01-26 |
| CBL-E-0047 | DC-East Virgin | MM Fiber OM4 | IDF-E-05 | RE-23/P37 | 25 | Pass | 2019-10-21 |
| CBL-E-0048 | DC-East Virgin | Cat 6A STP | IDF-E-05 | RE-24/P24 | 78 | Pass | 2018-02-03 |
| CBL-E-0049 | DC-East Virgin | MM Fiber OM4 | IDF-E-05 | RE-25/P47 | 24 | Pass | 2022-02-27 |
| CBL-E-0050 | DC-East Virgin | MM Fiber OM4 | IDF-E-06 | RE-01/P27 | 4 | Pass | 2021-02-28 |
| CBL-W-0001 | DC-West Oregon | MM Fiber OM4 | MDF-W | RW-02/P18 | 60 | Pass | 2023-03-20 |
| CBL-W-0002 | DC-West Oregon | Cat 6A STP | MDF-W | RW-03/P27 | 42 | Pass | 2019-11-28 |
| CBL-W-0003 | DC-West Oregon | SM Fiber | MDF-W | RW-04/P42 | 21 | Pass | 2019-03-20 |

| Cable ID | DC Location | Type | From | To | Length m | Test Result | Install Date |
|---|---|---|---|---|---|---|---|
| CBL-W-0004 | DC-West Oregon | SM Fiber | MDF-W | RW-05/P38 | 64 | Pass | 2019-03-15 |
| CBL-W-0005 | DC-West Oregon | Serial Conso | MDF-W | RW-06/P7 | 6 | Pass | 2018-09-03 |
| CBL-W-0006 | DC-West Oregon | SM Fiber | IDF-W-01 | RW-07/P2 | 20 | Pass | 2020-12-01 |
| CBL-W-0007 | DC-West Oregon | Cat 6A STP | IDF-W-01 | RW-08/P48 | 59 | Pass | 2019-07-30 |
| CBL-W-0008 | DC-West Oregon | Serial Conso | IDF-W-01 | RW-09/P34 | 48 | Pass | 2022-04-16 |
| CBL-W-0009 | DC-West Oregon | Cat 6A UTP | IDF-W-01 | RW-10/P42 | 29 | Pass | 2020-05-23 |
| CBL-W-0010 | DC-West Oregon | SM Fiber | IDF-W-02 | RW-11/P22 | 20 | Pass | 2021-04-26 |
| CBL-W-0011 | DC-West Oregon | MM Fiber OM4 | IDF-W-02 | RW-12/P47 | 58 | Pass | 2021-04-29 |
| CBL-W-0012 | DC-West Oregon | SM Fiber | IDF-W-02 | RW-13/P30 | 50 | Pass | 2019-09-19 |
| CBL-W-0013 | DC-West Oregon | Cat 6A UTP | IDF-W-02 | RW-14/P8 | 75 | Pass | 2021-03-03 |
| CBL-W-0014 | DC-West Oregon | Serial Conso | IDF-W-02 | RW-15/P9 | 44 | Pass | 2023-08-14 |
| CBL-W-0015 | DC-West Oregon | MM Fiber OM4 | IDF-W-02 | RW-16/P37 | 11 | Pass | 2023-08-09 |
| CBL-W-0016 | DC-West Oregon | Cat 6A UTP | IDF-W-02 | RW-17/P33 | 63 | Pass | 2019-09-30 |
| CBL-W-0017 | DC-West Oregon | SM Fiber | IDF-W-02 | RW-18/P43 | 42 | Pass | 2021-12-11 |
| CBL-W-0018 | DC-West Oregon | SM Fiber | IDF-W-02 | RW-19/P39 | 40 | Pass | 2022-12-04 |
| CBL-W-0019 | DC-West Oregon | Cat 6A UTP | IDF-W-02 | RW-20/P38 | 25 | Pass | 2024-06-22 |
| CBL-W-0020 | DC-West Oregon | MM Fiber OM4 | IDF-W-03 | RW-21/P47 | 14 | Pass | 2024-01-06 |
| CBL-W-0021 | DC-West Oregon | Cat 6A STP | IDF-W-03 | RW-22/P24 | 32 | Pass | 2024-06-10 |
| CBL-W-0022 | DC-West Oregon | MM Fiber OM4 | IDF-W-03 | RW-23/P39 | 54 | Pass | 2018-02-16 |
| CBL-W-0023 | DC-West Oregon | Cat 6A UTP | IDF-W-03 | RW-24/P18 | 62 | Pass | 2020-06-10 |
| CBL-W-0024 | DC-West Oregon | MM Fiber OM4 | IDF-W-03 | RW-25/P16 | 77 | Pass | 2022-02-17 |
| CBL-W-0025 | DC-West Oregon | Cat 6A UTP | IDF-W-03 | RW-01/P32 | 64 | Pass | 2018-08-08 |
| CBL-W-0026 | DC-West Oregon | SM Fiber | IDF-W-03 | RW-02/P1 | 32 | Pass | 2024-04-22 |
| CBL-W-0027 | DC-West Oregon | Cat 6A UTP | IDF-W-03 | RW-03/P34 | 10 | Pass | 2023-07-09 |
| CBL-W-0028 | DC-West Oregon | MM Fiber OM4 | IDF-W-03 | RW-04/P27 | 32 | Pass | 2018-06-09 |
| CBL-W-0029 | DC-West Oregon | MM Fiber OM4 | IDF-W-03 | RW-05/P20 | 43 | Pass | 2023-08-06 |
| CBL-W-0030 | DC-West Oregon | SM Fiber | IDF-W-04 | RW-06/P30 | 7 | Pass | 2023-04-28 |
| CBL-W-0031 | DC-West Oregon | Cat 6A UTP | IDF-W-04 | RW-07/P44 | 54 | Pass | 2023-12-09 |
| CBL-W-0032 | DC-West Oregon | Cat 6A UTP | IDF-W-04 | RW-08/P21 | 88 | Pass | 2020-01-12 |
| CBL-W-0033 | DC-West Oregon | SM Fiber | IDF-W-04 | RW-09/P43 | 12 | Pass | 2021-08-16 |
| CBL-W-0034 | DC-West Oregon | MM Fiber OM4 | IDF-W-04 | RW-10/P47 | 70 | Pass | 2023-04-10 |
| CBL-W-0035 | DC-West Oregon | Cat 6A UTP | IDF-W-04 | RW-11/P11 | 82 | Pass | 2024-04-28 |
| CBL-W-0036 | DC-West Oregon | MM Fiber OM4 | IDF-W-04 | RW-12/P16 | 17 | Pass | 2020-12-20 |
| CBL-W-0037 | DC-West Oregon | SM Fiber | IDF-W-04 | RW-13/P4 | 19 | Pass | 2020-12-23 |
| CBL-W-0038 | DC-West Oregon | MM Fiber OM4 | IDF-W-04 | RW-14/P8 | 48 | Marginal | 2019-04-18 |
| CBL-W-0039 | DC-West Oregon | MM Fiber OM4 | IDF-W-04 | RW-15/P39 | 55 | Pass | 2020-11-22 |
| CBL-W-0040 | DC-West Oregon | SM Fiber | IDF-W-05 | RW-16/P19 | 80 | Pass | 2021-03-08 |
| CBL-C-0001 | DC-Central Ill | Serial Conso | MDF-C | RC-02/P1 | 70 | Pass | 2020-03-23 |
| CBL-C-0002 | DC-Central Ill | MM Fiber OM4 | MDF-C | RC-03/P46 | 12 | Pass | 2018-11-16 |
| CBL-C-0003 | DC-Central Ill | Cat 6A UTP | MDF-C | RC-04/P15 | 11 | Pass | 2021-12-23 |
| CBL-C-0004 | DC-Central Ill | SM Fiber | MDF-C | RC-05/P47 | 33 | Pass | 2023-11-29 |
| CBL-C-0005 | DC-Central Ill | Serial Conso | MDF-C | RC-06/P11 | 16 | Pass | 2023-06-12 |
| CBL-C-0006 | DC-Central Ill | Cat 6A STP | IDF-C-01 | RC-07/P17 | 52 | Pass | 2020-04-18 |
| CBL-C-0007 | DC-Central Ill | Serial Conso | IDF-C-01 | RC-08/P46 | 89 | Pass | 2019-07-12 |
| CBL-C-0008 | DC-Central Ill | Serial Conso | IDF-C-01 | RC-09/P2 | 17 | Pass | 2021-01-16 |
| CBL-C-0009 | DC-Central Ill | SM Fiber | IDF-C-01 | RC-10/P40 | 59 | Pass | 2023-07-28 |
| CBL-C-0010 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-11/P38 | 28 | Pass | 2021-01-09 |
| CBL-C-0011 | DC-Central Ill | Serial Conso | IDF-C-02 | RC-12/P10 | 87 | Pass | 2020-12-03 |
| CBL-C-0012 | DC-Central Ill | Serial Conso | IDF-C-02 | RC-13/P15 | 46 | Pass | 2021-11-15 |
| CBL-C-0013 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-14/P3 | 57 | Pass | 2018-12-12 |
| CBL-C-0014 | DC-Central Ill | SM Fiber | IDF-C-02 | RC-15/P15 | 8 | Pass | 2022-03-12 |
| CBL-C-0015 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-16/P15 | 67 | Pass | 2020-07-19 |
| CBL-C-0016 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-17/P36 | 66 | Pass | 2019-06-30 |
| CBL-C-0017 | DC-Central Ill | MM Fiber OM4 | IDF-C-02 | RC-18/P17 | 16 | Pass | 2023-03-14 |
| CBL-C-0018 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-19/P41 | 35 | Pass | 2018-07-08 |
| CBL-C-0019 | DC-Central Ill | Cat 6A UTP | IDF-C-02 | RC-20/P45 | 62 | Pass | 2023-03-04 |
| CBL-C-0020 | DC-Central Ill | Cat 6A UTP | IDF-C-03 | RC-21/P7 | 42 | Pass | 2020-04-18 |
| CBL-C-0021 | DC-Central Ill | MM Fiber OM4 | IDF-C-03 | RC-22/P4 | 47 | Pass | 2021-07-20 |
| CBL-C-0022 | DC-Central Ill | MM Fiber OM4 | IDF-C-03 | RC-23/P34 | 16 | Pass | 2018-12-12 |
| CBL-C-0023 | DC-Central Ill | SM Fiber | IDF-C-03 | RC-24/P43 | 52 | Pass | 2023-05-23 |
| CBL-C-0024 | DC-Central Ill | Serial Conso | IDF-C-03 | RC-25/P7 | 9 | Pass | 2020-08-05 |
| CBL-C-0025 | DC-Central Ill | MM Fiber OM4 | IDF-C-03 | RC-01/P25 | 54 | Pass | 2023-03-31 |
| CBL-C-0026 | DC-Central Ill | SM Fiber | IDF-C-03 | RC-02/P7 | 13 | Pass | 2023-01-06 |
| CBL-C-0027 | DC-Central Ill | Serial Conso | IDF-C-03 | RC-03/P48 | 34 | Pass | 2019-04-29 |

| Cable ID | DC Location | Type | From | To | Length m | Test Result | Install Date |
|---|---|---|---|---|---|---|---|
| CBL-C-0028 | DC-Central III | Serial Conso | IDF-C-03 | RC-04/P9 | 31 | Pass | 2019-07-02 |
| CBL-C-0029 | DC-Central III | MM Fiber OM4 | IDF-C-03 | RC-05/P2 | 58 | Pass | 2022-01-07 |
| CBL-C-0030 | DC-Central III | SM Fiber | IDF-C-04 | RC-06/P31 | 18 | Pass | 2020-03-17 |

# Appendix G: Fire Suppression and Safety Systems

This appendix documents the fire suppression systems installed at all three datacenter facilities, including clean agent systems (FM-200/Novec 1230), pre-action sprinkler systems, and VESDA (Very Early Smoke Detection Apparatus) configurations.

All fire suppression systems are tested annually by a certified third-party contractor. The most recent full system test was conducted in September 2024 for all three facilities. Individual zone discharge tests are conducted quarterly with the most recent Q4 test completed on November 8, 2024.

Detection zones are mapped to server room zones with a typical coverage of 4-6 racks per detection zone. Cross-zone detection requires activation in two adjacent zones before system discharge to prevent false activations. Agent hold time is certified at minimum 10 minutes for all rooms at current seal integrity levels.

| Zone ID | DC Location | System Type | Agent | Racks Covered | Last Test | Test Result | Next Due |
|---|---|---|---|---|---|---|---|
| FZ-E-01 | DC-East Virgin | Clean Agent | Novec 1230 | 4 racks | 2024-11-23 | Pass | 2025-02-21 |
| FZ-E-02 | DC-East Virgin | Clean Agent | Novec 1230 | 7 racks | 2024-11-25 | Pass | 2025-02-23 |
| FZ-E-03 | DC-East Virgin | Clean Agent | Novec 1230 | 7 racks | 2024-11-17 | Pass | 2025-02-15 |
| FZ-E-04 | DC-East Virgin | Clean Agent | Novec 1230 | 4 racks | 2024-11-15 | Marginal | 2025-02-13 |
| FZ-E-05 | DC-East Virgin | Clean Agent | Novec 1230 | 7 racks | 2024-10-17 | Pass | 2025-01-15 |
| FZ-E-06 | DC-East Virgin | Clean Agent | Novec 1230 | 6 racks | 2024-11-05 | Pass | 2025-02-03 |
| FZ-E-07 | DC-East Virgin | Clean Agent | Novec 1230 | 8 racks | 2024-10-21 | Pass | 2025-01-19 |
| FZ-E-08 | DC-East Virgin | Clean Agent | Novec 1230 | 8 racks | 2024-09-19 | Pass | 2024-12-18 |
| FZ-E-09 | DC-East Virgin | Clean Agent | Novec 1230 | 6 racks | 2024-11-04 | Pass | 2025-02-02 |
| FZ-E-10 | DC-East Virgin | Clean Agent | Novec 1230 | 4 racks | 2024-09-17 | Pass | 2024-12-16 |
| FZ-E-11 | DC-East Virgin | Pre-Action Spr | Water | 4 racks | 2024-09-15 | Pass | 2024-12-14 |
| FZ-E-12 | DC-East Virgin | Pre-Action Spr | Water | 8 racks | 2024-10-18 | Pass | 2025-01-16 |
| FZ-W-01 | DC-West Oregon | Clean Agent | Novec 1230 | 6 racks | 2024-09-12 | Pass | 2024-12-11 |
| FZ-W-02 | DC-West Oregon | Clean Agent | Novec 1230 | 4 racks | 2024-09-16 | Marginal | 2024-12-15 |
| FZ-W-03 | DC-West Oregon | Clean Agent | Novec 1230 | 5 racks | 2024-11-19 | Pass | 2025-02-17 |
| FZ-W-04 | DC-West Oregon | Clean Agent | Novec 1230 | 6 racks | 2024-10-11 | Pass | 2025-01-09 |
| FZ-W-05 | DC-West Oregon | Clean Agent | Novec 1230 | 7 racks | 2024-09-24 | Pass | 2024-12-23 |
| FZ-W-06 | DC-West Oregon | Clean Agent | Novec 1230 | 4 racks | 2024-09-15 | Pass | 2024-12-14 |
| FZ-W-07 | DC-West Oregon | Clean Agent | Novec 1230 | 7 racks | 2024-09-18 | Pass | 2024-12-17 |
| FZ-W-08 | DC-West Oregon | Clean Agent | Novec 1230 | 6 racks | 2024-11-06 | Pass | 2025-02-04 |
| FZ-W-09 | DC-West Oregon | Pre-Action Spr | Water | 4 racks | 2024-09-12 | Pass | 2024-12-11 |
| FZ-W-10 | DC-West Oregon | Pre-Action Spr | Water | 8 racks | 2024-09-30 | Pass | 2024-12-29 |
| FZ-C-01 | DC-Central Ill | Clean Agent | Novec 1230 | 7 racks | 2024-11-10 | Pass | 2025-02-08 |
| FZ-C-02 | DC-Central Ill | Clean Agent | Novec 1230 | 8 racks | 2024-10-11 | Pass | 2025-01-09 |
| FZ-C-03 | DC-Central Ill | Clean Agent | Novec 1230 | 8 racks | 2024-09-19 | Pass | 2024-12-18 |
| FZ-C-04 | DC-Central Ill | Clean Agent | Novec 1230 | 6 racks | 2024-11-27 | Marginal | 2025-02-25 |
| FZ-C-05 | DC-Central Ill | Clean Agent | Novec 1230 | 6 racks | 2024-09-20 | Marginal | 2024-12-19 |
| FZ-C-06 | DC-Central Ill | Clean Agent | Novec 1230 | 4 racks | 2024-11-28 | Pass | 2025-02-26 |
| FZ-C-07 | DC-Central Ill | Pre-Action Spr | Water | 4 racks | 2024-10-16 | Pass | 2025-01-14 |
| FZ-C-08 | DC-Central Ill | Pre-Action Spr | Water | 7 racks | 2024-09-07 | Pass | 2024-12-06 |

# Appendix H: Physical Security Access Log Summary - Q4 2024

This appendix summarizes physical access events to datacenter facilities during Q4 2024. Access is controlled via multi-factor authentication (badge + biometric) at all entry points. Visitor access requires pre-authorization and escort at all times. The data below is aggregated from the facility access control system and CCTV event logs.

Access anomalies requiring investigation are flagged in the final column. Anomalies include badge tailgating events, failed biometric attempts exceeding 3 in succession, access outside of scheduled maintenance windows, and badge usage at unusual hours (between 22:00 and 06:00 local time).

| Date | DC Location | Access Point | Personnel | Badge ID | Entry Time | Exit Time | Anomaly |
|---|---|---|---|---|---|---|---|
| 2024-10-28 | DC-East Virgin | Network Room | K. Smith | BDG-5228 | 13:14 | 21:14 | None |
| 2024-12-15 | DC-Central Ill | Server Room A | S. Vance | BDG-7522 | 19:38 | 20:53 | None |
| 2024-11-06 | DC-West Oregon | Network Room | V. Patel | BDG-8530 | 16:51 | 20:13 | None |
| 2024-11-24 | DC-West Oregon | Server Room B | A. Davis | BDG-3086 | 14:08 | 19:36 | None |
| 2024-11-02 | DC-Central Ill | Loading Dock | G. Fischer | BDG-6147 | 09:24 | 13:31 | None |
| 2024-10-28 | DC-Central Ill | Server Room A | G. Fischer | BDG-2626 | 20:02 | 23:27 | None |
| 2024-11-01 | DC-West Oregon | Power Room | S. Vance | BDG-2732 | 08:56 | 09:54 | None |
| 2024-12-18 | DC-East Virgin | Loading Dock | G. Fischer | BDG-9794 | 20:07 | 23:53 | None |
| 2024-10-18 | DC-Central Ill | Server Room B | G. Fischer | BDG-6007 | 12:57 | 17:33 | None |
| 2024-12-11 | DC-West Oregon | Main Entrance | Vendor - Dell | BDG-8022 | 21:17 | 22:37 | None |
| 2024-12-12 | DC-West Oregon | Loading Dock | Vendor - HPE | BDG-3743 | 15:16 | 17:51 | None |
| 2024-12-01 | DC-Central Ill | Main Entrance | J. Miller | BDG-5747 | 12:00 | 19:28 | None |
| 2024-11-29 | DC-East Virgin | Loading Dock | Vendor - HPE | BDG-6694 | 19:17 | 22:45 | None |
| 2024-12-20 | DC-West Oregon | Power Room | A. Davis | BDG-5878 | 22:00 | 23:48 | None |
| 2024-12-03 | DC-West Oregon | Loading Dock | Vendor - HPE | BDG-1001 | 18:44 | 23:13 | None |
| 2024-12-13 | DC-West Oregon | Loading Dock | T. Kelly | BDG-6495 | 19:45 | 22:06 | None |
| 2024-10-16 | DC-West Oregon | Main Entrance | V. Patel | BDG-3061 | 17:15 | 18:07 | None |
| 2024-10-12 | DC-Central Ill | Network Room | Vendor - Dell | BDG-3858 | 06:42 | 09:55 | None |
| 2024-11-19 | DC-Central Ill | Server Room A | T. Kelly | BDG-4667 | 20:45 | 23:21 | None |
| 2024-11-18 | DC-East Virgin | Main Entrance | K. Smith | BDG-5129 | 22:39 | 23:56 | None |
| 2024-11-06 | DC-West Oregon | Power Room | A. Davis | BDG-9918 | 17:07 | 21:56 | None |
| 2024-12-11 | DC-West Oregon | Loading Dock | P. Murphy | BDG-4358 | 11:49 | 16:23 | None |
| 2024-11-23 | DC-East Virgin | Loading Dock | Vendor - HPE | BDG-7243 | 17:55 | 18:56 | None |
| 2024-10-20 | DC-West Oregon | Network Room | R. Brooks | BDG-1146 | 08:41 | 11:47 | None |
| 2024-10-25 | DC-East Virgin | Server Room A | K. Smith | BDG-7434 | 07:59 | 12:36 | None |
| 2024-10-04 | DC-Central Ill | Power Room | T. Kelly | BDG-7142 | 13:32 | 16:43 | None |
| 2024-11-17 | DC-Central Ill | Server Room B | V. Patel | BDG-9037 | 16:00 | 19:10 | None |
| 2024-10-03 | DC-Central Ill | Loading Dock | J. Miller | BDG-5415 | 20:38 | 23:42 | None |
| 2024-11-22 | DC-East Virgin | Loading Dock | E. Howard | BDG-9888 | 08:45 | 15:10 | None |
| 2024-10-29 | DC-West Oregon | Loading Dock | S. Vance | BDG-6581 | 11:46 | 19:44 | None |
| 2024-10-16 | DC-Central Ill | Main Entrance | S. Vance | BDG-8365 | 09:36 | 11:36 | None |
| 2024-11-08 | DC-East Virgin | Main Entrance | R. Brooks | BDG-9548 | 11:09 | 19:32 | None |
| 2024-12-05 | DC-Central Ill | Main Entrance | S. Vance | BDG-5778 | 14:38 | 16:13 | Tailgating |
| 2024-11-03 | DC-West Oregon | Loading Dock | V. Patel | BDG-4947 | 17:21 | 20:08 | None |
| 2024-10-15 | DC-Central Ill | Server Room B | J. Miller | BDG-7361 | 13:03 | 17:02 | None |
| 2024-10-24 | DC-East Virgin | Power Room | E. Howard | BDG-9052 | 08:25 | 13:47 | None |
| 2024-11-22 | DC-West Oregon | Server Room B | J. Miller | BDG-1324 | 06:35 | 13:30 | None |
| 2024-10-28 | DC-West Oregon | Server Room B | A. Davis | BDG-3890 | 15:22 | 19:06 | None |
| 2024-12-02 | DC-West Oregon | Server Room A | E. Howard | BDG-8551 | 07:36 | 14:01 | None |
| 2024-10-21 | DC-Central Ill | Server Room B | A. Davis | BDG-3758 | 19:30 | 21:38 | None |
| 2024-11-10 | DC-East Virgin | Loading Dock | A. Davis | BDG-5135 | 20:01 | 23:17 | None |
| 2024-12-09 | DC-East Virgin | Main Entrance | T. Kelly | BDG-2906 | 21:17 | 23:43 | None |
| 2024-12-08 | DC-West Oregon | Power Room | J. Miller | BDG-7022 | 15:19 | 20:30 | None |
| 2024-10-01 | DC-East Virgin | Server Room A | Vendor - HPE | BDG-5131 | 22:27 | 23:42 | None |
| 2024-11-18 | DC-Central Ill | Power Room | Vendor - Dell | BDG-6462 | 18:59 | 20:25 | Tailgating |
| 2024-10-12 | DC-West Oregon | Loading Dock | P. Murphy | BDG-5907 | 08:35 | 09:29 | None |
| 2024-11-11 | DC-East Virgin | Server Room A | S. Vance | BDG-6993 | 10:18 | 14:16 | None |
| 2024-10-18 | DC-East Virgin | Network Room | E. Howard | BDG-5143 | 16:12 | 17:32 | None |
| 2024-12-12 | DC-Central Ill | Loading Dock | Vendor - HPE | BDG-7064 | 19:05 | 23:18 | None |
| 2024-12-04 | DC-Central Ill | Loading Dock | P. Murphy | BDG-5094 | 18:49 | 23:35 | None |
| 2024-11-24 | DC-East Virgin | Main Entrance | K. Smith | BDG-9991 | 09:09 | 12:52 | None |
| 2024-10-27 | DC-East Virgin | Server Room A | S. Vance | BDG-8126 | 12:37 | 19:00 | None |
| 2024-10-31 | DC-Central Ill | Server Room B | E. Howard | BDG-2136 | 21:09 | 22:03 | None |

| Date | DC Location | Access Point | Personnel | Badge ID | Entry Time | Exit Time | Anomaly |
|---|---|---|---|---|---|---|---|
| 2024-10-23 | DC-West Oregon | Power Room | S. Vance | BDG-6464 | 07:58 | 11:44 | None |
| 2024-10-25 | DC-West Oregon | Power Room | K. Smith | BDG-6881 | 15:02 | 20:38 | None |
| 2024-10-31 | DC-Central Ill | Network Room | T. Kelly | BDG-5531 | 13:36 | 20:08 | None |
| 2024-12-02 | DC-East Virgin | Power Room | V. Patel | BDG-2126 | 21:40 | 23:23 | None |
| 2024-10-24 | DC-Central Ill | Server Room A | V. Patel | BDG-1292 | 09:06 | 16:56 | Tailgating |
| 2024-12-08 | DC-Central Ill | Server Room A | V. Patel | BDG-1199 | 17:13 | 18:19 | None |
| 2024-10-07 | DC-Central Ill | Main Entrance | S. Vance | BDG-1545 | 18:14 | 21:48 | None |
| 2024-11-28 | DC-Central Ill | Loading Dock | E. Howard | BDG-8759 | 21:41 | 22:29 | None |
| 2024-12-10 | DC-Central Ill | Power Room | V. Patel | BDG-5743 | 20:01 | 23:06 | None |
| 2024-11-05 | DC-West Oregon | Server Room A | T. Kelly | BDG-1213 | 12:29 | 15:58 | None |
| 2024-11-27 | DC-East Virgin | Power Room | K. Smith | BDG-5551 | 21:35 | 23:18 | None |
| 2024-11-20 | DC-Central Ill | Network Room | K. Smith | BDG-8061 | 12:55 | 17:35 | None |
| 2024-11-04 | DC-Central Ill | Server Room A | K. Smith | BDG-1250 | 09:48 | 16:37 | None |
| 2024-11-06 | DC-West Oregon | Main Entrance | V. Patel | BDG-8182 | 10:26 | 12:25 | None |
| 2024-12-20 | DC-West Oregon | Server Room A | V. Patel | BDG-9245 | 12:39 | 17:24 | None |
| 2024-10-06 | DC-Central Ill | Power Room | T. Kelly | BDG-4520 | 13:10 | 19:54 | None |
| 2024-11-12 | DC-East Virgin | Network Room | A. Davis | BDG-1518 | 14:23 | 22:32 | None |
| 2024-10-15 | DC-East Virgin | Power Room | A. Davis | BDG-5825 | 08:16 | 16:03 | None |
| 2024-12-03 | DC-West Oregon | Main Entrance | T. Kelly | BDG-8112 | 14:21 | 16:40 | None |
| 2024-11-09 | DC-West Oregon | Network Room | R. Brooks | BDG-5650 | 16:12 | 20:14 | After-hours |
| 2024-12-01 | DC-East Virgin | Main Entrance | K. Smith | BDG-2468 | 11:39 | 16:56 | None |
| 2024-11-11 | DC-Central Ill | Network Room | T. Kelly | BDG-8175 | 16:48 | 21:06 | None |
| 2024-11-18 | DC-Central Ill | Main Entrance | A. Davis | BDG-4907 | 15:41 | 18:55 | None |
| 2024-12-07 | DC-East Virgin | Server Room A | K. Smith | BDG-4856 | 16:15 | 21:43 | None |
| 2024-11-08 | DC-Central Ill | Network Room | A. Davis | BDG-7447 | 17:19 | 20:33 | None |
| 2024-11-02 | DC-West Oregon | Network Room | R. Brooks | BDG-7809 | 19:31 | 23:48 | None |
| 2024-11-03 | DC-Central Ill | Loading Dock | Vendor - Dell | BDG-8304 | 13:57 | 17:17 | None |
| 2024-10-25 | DC-East Virgin | Power Room | Vendor - HPE | BDG-9502 | 10:47 | 16:31 | None |
| 2024-12-19 | DC-West Oregon | Server Room A | E. Howard | BDG-9508 | 07:14 | 12:48 | None |
| 2024-10-23 | DC-East Virgin | Loading Dock | G. Fischer | BDG-6166 | 08:29 | 14:40 | None |
| 2024-10-26 | DC-East Virgin | Server Room A | A. Davis | BDG-5843 | 13:07 | 17:59 | None |
| 2024-12-08 | DC-East Virgin | Network Room | T. Kelly | BDG-2973 | 12:28 | 16:31 | None |
| 2024-10-31 | DC-East Virgin | Power Room | E. Howard | BDG-4456 | 14:39 | 19:31 | None |
| 2024-10-28 | DC-East Virgin | Network Room | K. Smith | BDG-2957 | 09:14 | 13:02 | Failed bio x3 |
| 2024-10-26 | DC-West Oregon | Server Room B | T. Kelly | BDG-7741 | 06:05 | 08:20 | None |
| 2024-10-30 | DC-Central Ill | Power Room | T. Kelly | BDG-4537 | 17:10 | 19:42 | None |
| 2024-11-30 | DC-Central Ill | Main Entrance | P. Murphy | BDG-1834 | 17:09 | 23:39 | None |
| 2024-10-09 | DC-East Virgin | Main Entrance | R. Brooks | BDG-4986 | 15:14 | 19:42 | None |
| 2024-11-07 | DC-Central Ill | Server Room B | S. Vance | BDG-7196 | 14:48 | 22:00 | Tailgating |
| 2024-12-02 | DC-East Virgin | Server Room B | P. Murphy | BDG-5874 | 13:32 | 19:48 | None |
| 2024-10-28 | DC-Central Ill | Main Entrance | K. Smith | BDG-2918 | 07:44 | 15:18 | Failed bio x3 |
| 2024-12-17 | DC-West Oregon | Power Room | T. Kelly | BDG-3741 | 10:45 | 14:39 | None |
| 2024-11-05 | DC-Central Ill | Server Room A | V. Patel | BDG-6016 | 06:52 | 12:29 | None |
| 2024-11-15 | DC-Central Ill | Server Room A | S. Vance | BDG-9757 | 20:45 | 22:25 | None |
| 2024-10-22 | DC-West Oregon | Server Room A | J. Miller | BDG-2318 | 20:28 | 23:12 | None |
| 2024-11-18 | DC-Central Ill | Power Room | Vendor - Dell | BDG-1409 | 11:07 | 19:06 | None |
| 2024-12-01 | DC-East Virgin | Server Room A | S. Vance | BDG-8732 | 14:59 | 15:53 | Failed bio x3 |
| 2024-10-16 | DC-Central Ill | Loading Dock | A. Davis | BDG-8489 | 08:41 | 12:02 | After-hours |
| 2024-11-12 | DC-East Virgin | Server Room B | A. Davis | BDG-1818 | 20:11 | 23:21 | None |
| 2024-11-06 | DC-Central Ill | Power Room | P. Murphy | BDG-6380 | 07:02 | 10:14 | Tailgating |
| 2024-11-16 | DC-Central Ill | Network Room | G. Fischer | BDG-5657 | 14:37 | 20:28 | None |
| 2024-11-01 | DC-East Virgin | Power Room | S. Vance | BDG-2837 | 08:19 | 13:29 | After-hours |
| 2024-11-02 | DC-East Virgin | Loading Dock | R. Brooks | BDG-4986 | 21:06 | 22:26 | None |
| 2024-10-16 | DC-East Virgin | Loading Dock | V. Patel | BDG-3923 | 15:46 | 19:17 | None |
| 2024-11-28 | DC-East Virgin | Loading Dock | G. Fischer | BDG-4242 | 16:52 | 21:04 | None |
| 2024-10-05 | DC-West Oregon | Main Entrance | G. Fischer | BDG-5966 | 17:58 | 23:38 | None |
| 2024-11-29 | DC-East Virgin | Power Room | Vendor - HPE | BDG-9957 | 16:18 | 20:16 | None |
| 2024-10-26 | DC-Central Ill | Loading Dock | K. Smith | BDG-5318 | 19:45 | 21:16 | None |
| 2024-10-28 | DC-West Oregon | Power Room | Vendor - HPE | BDG-2913 | 11:22 | 13:55 | None |
| 2024-11-21 | DC-East Virgin | Network Room | S. Vance | BDG-3451 | 17:46 | 23:29 | None |
| 2024-10-27 | DC-Central Ill | Main Entrance | E. Howard | BDG-3294 | 13:59 | 20:09 | Tailgating |
| 2024-10-25 | DC-East Virgin | Server Room B | T. Kelly | BDG-9860 | 15:40 | 19:14 | None |
| 2024-11-25 | DC-East Virgin | Server Room B | E. Howard | BDG-7622 | 10:35 | 17:04 | None |
| 2024-11-20 | DC-West Oregon | Power Room | J. Miller | BDG-1871 | 13:49 | 14:39 | None |

| Date | DC Location | Access Point | Personnel | Badge ID | Entry Time | Exit Time | Anomaly |
| --- | --- | --- | --- | --- | --- | --- | --- |
| 2024-11-08 | DC-East Virgin | Server Room B | K. Smith | BDG-6081 | 14:56 | 19:35 | Tailgating |
| 2024-11-22 | DC-West Oregon | Power Room | T. Kelly | BDG-4187 | 22:02 | 23:47 | None |
| 2024-10-09 | DC-Central Ill | Server Room B | Vendor - HPE | BDG-8431 | 14:03 | 21:53 | None |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2024-12-15 | 1.43 | 21.2 C | 1.35 | 19.4 C | 1.55 | 21.6 C |
| 2024-12-16 | 1.46 | 22.3 C | 1.35 | 19.6 C | 1.55 | 21.7 C |
| 2024-12-17 | 1.46 | 21.0 C | 1.33 | 20.2 C | 1.57 | 21.6 C |
| 2024-12-18 | 1.44 | 20.6 C | 1.34 | 20.3 C | 1.55 | 22.3 C |
| 2024-12-19 | 1.44 | 21.5 C | 1.35 | 20.3 C | 1.54 | 21.1 C |
| 2024-12-20 | 1.46 | 21.7 C | 1.35 | 19.4 C | 1.56 | 21.5 C |
| 2024-12-21 | 1.46 | 21.6 C | 1.35 | 20.4 C | 1.54 | 21.9 C |
| 2024-12-22 | 1.43 | 21.3 C | 1.33 | 20.2 C | 1.53 | 22.3 C |
| 2024-12-23 | 1.44 | 22.5 C | 1.36 | 20.0 C | 1.57 | 22.3 C |
| 2024-12-24 | 1.46 | 20.8 C | 1.33 | 19.8 C | 1.57 | 21.5 C |
| 2024-12-25 | 1.47 | 21.3 C | 1.34 | 19.7 C | 1.56 | 21.6 C |
| 2024-12-26 | 1.44 | 22.4 C | 1.35 | 19.4 C | 1.56 | 21.1 C |
| 2024-12-27 | 1.44 | 22.1 C | 1.34 | 19.6 C | 1.55 | 22.8 C |
| 2024-12-28 | 1.46 | 22.5 C | 1.34 | 20.0 C | 1.56 | 21.5 C |
| 2024-12-29 | 1.44 | 21.7 C | 1.34 | 20.4 C | 1.56 | 22.1 C |
| 2024-12-30 | 1.46 | 21.4 C | 1.36 | 19.4 C | 1.55 | 21.0 C |
| 2024-12-31 | 1.44 | 20.6 C | 1.34 | 20.1 C | 1.56 | 22.8 C |

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